

Mapping the Consumption Gap:
Planned Data Center Capacity versus Electricity Supply in
California

September 22, 2026

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Abstract

Chapter 1

Introduction

Chapter 2

Current Status of Data Centers and the Grid in California

This chapter establishes the baseline that the gap analysis in later chapters is measured against. It answers two questions: what the CAISO grid looks like hour by hour today, and how much electricity California’s existing data centers already use. Every number is computed from the raw files listed in `data/raw/manifest.csv`; the notebook `notebooks/01_baseline_grid.ipynb` reproduces each table and figure.

2.1 Grid baseline: CAISO demand, net load and imports, 2019–2025

Hourly demand, generation by fuel and net interchange for the CAISO balancing authority come from the EIA-930 Hourly Electric Grid Monitor, using EIA’s “Adjusted” series [U.S. Energy Information Administration, 2026d]. Net load is demand minus utility-scale solar and wind. Two definitional points matter. First, EIA-930 demand for CISO includes battery charging and pumping load, whereas CAISO’s own reported demand excludes charging; in 2025 the hourly gap between the two series tracks battery charging almost one for one (slope 0.91, correlation 0.87). Both definitions are kept in the processed table. Second, EIA’s series contains reporting artifacts, chiefly in early 2019, where demand falls implausibly below CAISO’s demand plus charging; those hours are nulled rather than corrected, leaving 8,435 valid hours in 2019 (325 missing, 3.7 percent) and at most 27 missing hours in any later year; annual energy is the mean of the valid hours times the calendar hours, which imputes the missing hours at the annual mean, and the duration curves and profiles below are drawn over valid hours only. Third, EIA-930 carries no CISO interchange at all, in any of its reported, imputed or adjusted variants, from July 2 to November 3, 2024, and no hydro for parts of 2019 and 2020; those hours are filled from CAISO’s own five-minute imports and hydro series through a per-year linear calibration fitted on the overlapping hours (correlation 0.986 or better; the fill log is `ch1_eia930_fill_log.csv`). Fourth, EIA warns that CAISO changed its data collection for natural gas in December 2023 and that the EIA-930 gas series is not comparable across that date [U.S. Energy Information Administration, 2025b]. The warning is confirmed here: through November 2023 EIA’s CISO gas generation matches CAISO’s own five-minute fuel mix to within 0.1 TWh a month, and from December 2023 it exceeds it by 0.7 to 1.8 TWh a month, 20 TWh in 2024 and 25 TWh in 2025 (`ch1_gas_series_comparability_monthly.csv`). Gas figures in Tables 2.1 and 2.2 are EIA’s series and must not be compared across that month.

Table 2.1: CAISO annual baseline from EIA-930 (Adjusted series), 2019–2025. Demand energy is the mean of valid hours times calendar hours. Storage charging is the estimated net charging of the CAISO battery fleet from the five-minute fuel mix. Import share is net imports divided by demand. Gas is EIA’s series, not comparable across December 2023.

Year	Demand (TWh)	Peak (GW)	Load factor	Peak net load (GW)	Min net load (GW)	Solar (TWh)	Wind (TWh)	Storage charging (TWh)	Net imports (TWh)	Imports (% of demand)
2019	215.8	43.8	0.56	40.5	5.2	27.7	15.9	0.1	53.8	25
2020	218.0	46.6	0.53	44.0	5.6	28.6	14.9	0.1	59.4	27
2021	219.2	43.6	0.57	40.5	3.0	32.8	18.3	0.9	54.5	25
2022	223.7	51.1	0.50	47.5	1.9	35.8	17.4	1.7	48.7	22
2023	218.2	44.0	0.57	41.8	1.7	37.2	16.4	4.4	28.8	13
2024	224.1	47.6	0.54	42.4	0.3	44.8	20.1	9.5	31.7	14
2025	224.1	43.9	0.58	39.8	-3.3	49.7	19.8	14.2	36.1	16

Table 2.1 shows a grid whose annual demand has been flat since 2019 while its composition changed. Demand ranged between 216 and 224 TWh in every year. Utility-scale solar output rose from 27.7 TWh to 49.7 TWh and the combined solar and wind share of demand went from a fifth to nearly a third. Net imports fell from a quarter of demand in 2019 to 13 percent in 2023 and 14 to 16 percent in 2024 and 2025. Battery charging, negligible in 2019, reached 14 TWh in 2025, and battery discharge peaked above 10 GW (Table 2.5). The hourly demand record remains 51.1 GW, set on September 6, 2022 at 18:00.

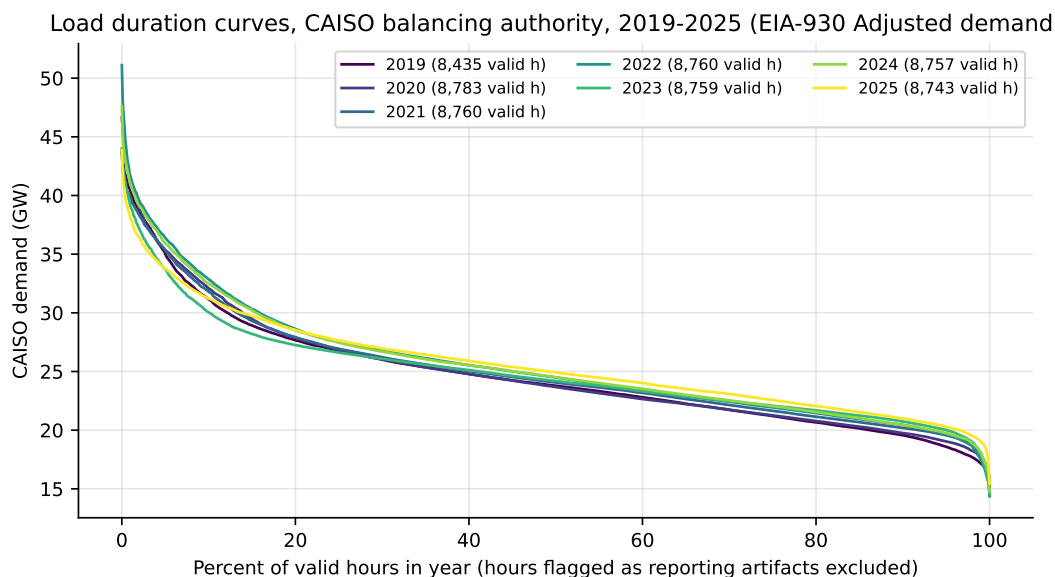


Figure 2.1: Load duration curves over valid hours (counts in the legend), CAISO, 2019–2025. Hours flagged as reporting artifacts are excluded and interchange and hydro gaps are filled from CAISO series. Source: EIA-930 Adjusted demand.

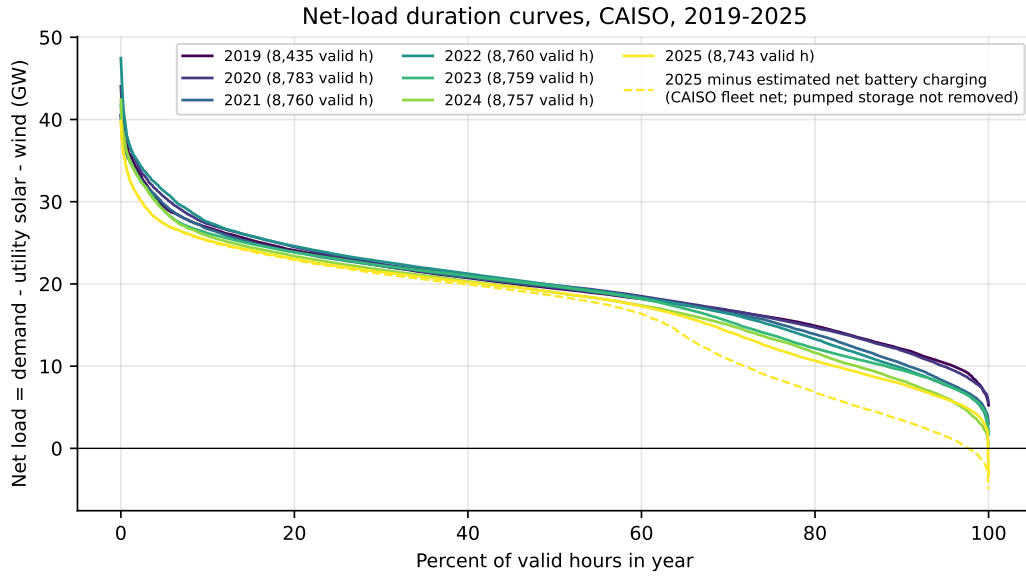


Figure 2.2: Net-load duration curves over valid hours, CAISO, 2019–2025. The dashed line subtracts the estimated net battery charging of the CAISO fleet from 2025 demand; it does not remove pumped storage and cannot recover gross charging in hours when part of the fleet discharges. Source: EIA-930, CAISO fuel mix.

Figures 2.1 and 2.2 make the contrast visible. The load duration curves are nearly indistinguishable across years. The net-load duration curves diverge at the low end: by 2024 and 2025 the lowest-net-load hours approach zero, and once battery charging, which is itself a response to the surplus, is removed from demand, utility solar and wind alone exceeded end-use demand in 80 hours of 2024 and 205 hours of 2025. This midday surplus is the physical basis for the flexibility question in Chapter 5.

Average daily profiles by season, CAISO: demand (top) and net load (bottom), 2019 / 2022 / 2025

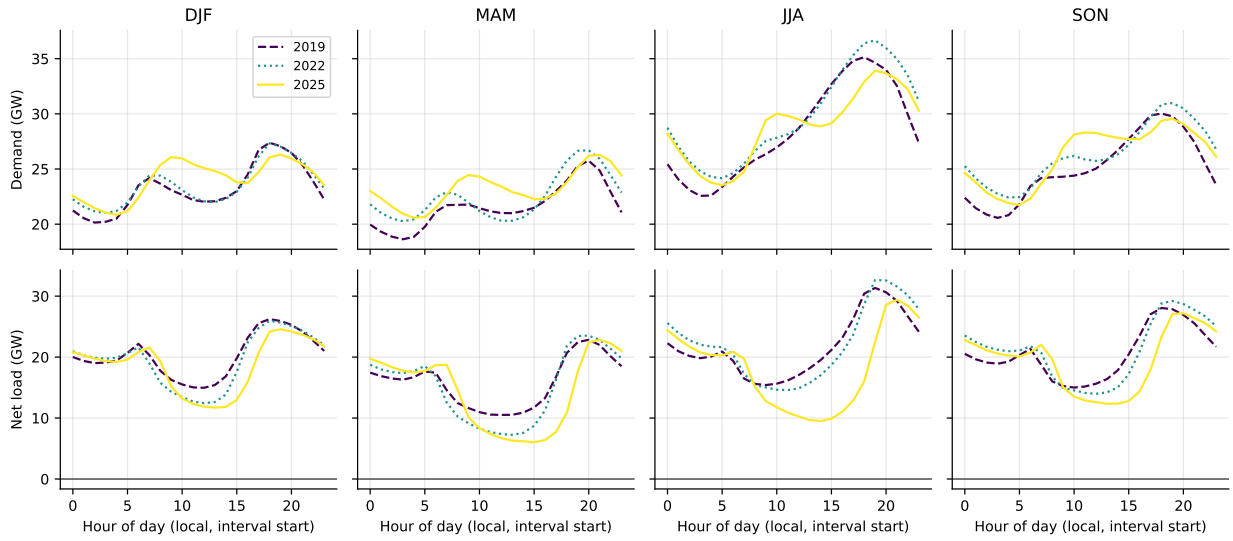


Figure 2.3: Average daily profiles by season: demand (top) and net load (bottom), 2019, 2022 and 2025. Means over valid hours; descriptive averages, not weather-normalised, and demand includes storage charging and pumping. Source: EIA-930.

The seasonal profiles in Figure 2.3 show the mechanism. Demand profiles are similar across years, with a modest midday rise in 2025 that reflects storage charging. Net-load profiles deepen at midday and steepen in the evening: in spring the average midday net load fell from about 10.5 GW in 2019 to about 6 GW in 2025 while the 20:00 value stayed near 22 GW, and in summer the midday trough fell from about 15 GW to under 10 GW.

Table 2.2: The 100 highest net-load hours of each year: when they occur and what serves them. Mean gas is EIA’s series (comparability note in the text). Source: EIA-930.

Year	Net load range (GW)	Distinct days	Jul-Sep (%)	17:00–21:00 (%)	Median start	Mean solar (GW)	Mean imports (GW)	Mean gas (GW)
2019	35.6–40.5	34	90	99	18:00	0.9	8.5	18.5
2020	36.4–44.0	30	92	86	18:00	1.2	7.4	22.6
2021	34.7–40.5	42	92	100	19:00	0.6	6.3	21.0
2022	36.6–47.5	27	95	82	19:00	1.2	7.4	22.5
2023	35.0–41.8	35	96	93	20:00	0.6	3.6	21.8
2024	34.7–42.4	37	91	79	20:00	0.9	4.7	20.9
2025	32.3–39.8	33	98	78	20:00	0.7	4.6	17.3

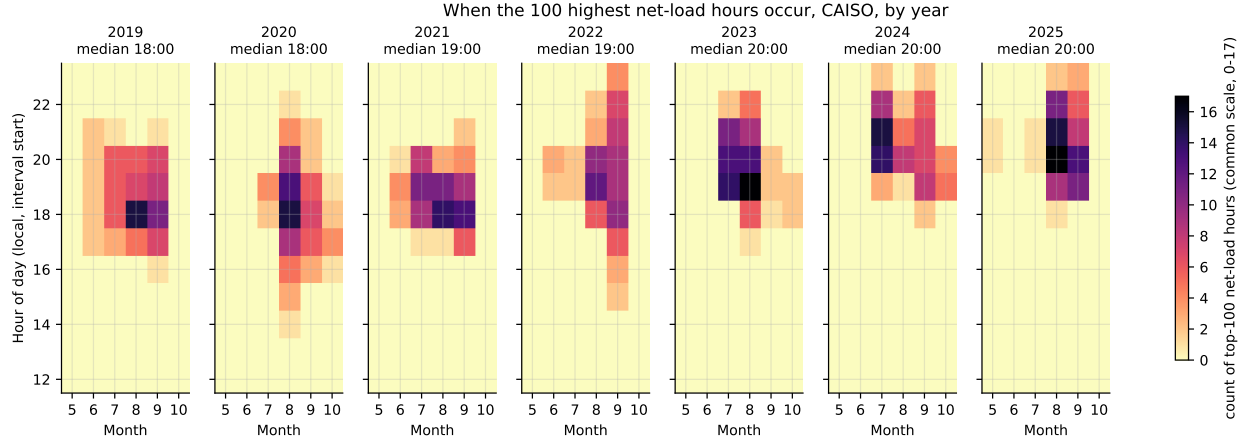


Figure 2.4: Month and hour of the 100 highest net-load hours, by year, on one common colour scale. Source: EIA-930.

Table 2.2 and Figure 2.4 identify the hours that set capacity needs. In every year at least 90 percent of the top 100 net-load hours fall in July through September and at least 78 percent between 17:00 and 21:00. Their median start time moved from 18:00 in 2019 and 2020 to 19:00 in 2021 and 2022 and to 20:00 from 2023 onward, the signature of solar pushing the net peak past sunset and of storage shaving the early evening. Solar contributes between 0.6 and 1.2 GW on average in these hours; gas contributes 17 to 23 GW and imports 4 to 9 GW. The implication for a flat load such as a data center is direct: each megawatt adds one for one to hours that solar does not serve, and the relevant capacity resources are storage, gas and imports.

2.2 Day-ahead prices and negative-price hours

Hourly day-ahead prices at the three investor-owned utility default load aggregation points come from CAISO OASIS [California Independent System Operator, 2026c]. OASIS retains roughly 39 months of data, so hourly prices run from July 2023; 2024 and 2025 are complete years and 2026 covers January 1 to September 22. EIA's daily ICE hub prices at NP15 and SP15 extend to 2015 but are daily block averages [U.S. Energy Information Administration, 2026e].

Table 2.3: Day-ahead LMP statistics at CAISO default load aggregation points. Source: CAISO OASIS PRC_LMP, DAM.

Node	Year	Hours	Mean	Median	P5	P95	Min	Hours < \$0	Hours ≤ \$5	Negative hours 10:00–16:00 (%)
(\$/MWh)										
PG&E DLAP	2024	8784	43.0	40.1	9.0	76.2	-33.1	190	327	88
PG&E DLAP	2025	8760	40.6	42.4	8.7	63.8	-36.8	218	337	94
PG&E DLAP	2026*	6359	31.5	31.9	0.4	60.1	-14.2	294	480	87
SCE DLAP	2024	8784	34.0	37.0	-20.0	67.7	-66.6	1131	1369	77
SCE DLAP	2025	8760	34.4	40.2	-8.3	60.5	-40.5	877	1109	78
SCE DLAP	2026*	6359	28.7	30.3	-9.0	64.8	-17.9	890	1123	72
SDG&E DLAP	2024	8784	36.2	38.2	-17.3	72.5	-63.1	989	1227	77
SDG&E DLAP	2025	8760	37.5	42.4	-6.8	65.8	-40.1	748	955	80
SDG&E DLAP	2026*	6359	32.3	33.1	-7.2	70.3	-17.2	706	950	74

* partial year: 2026 covers January 1 to September 22.

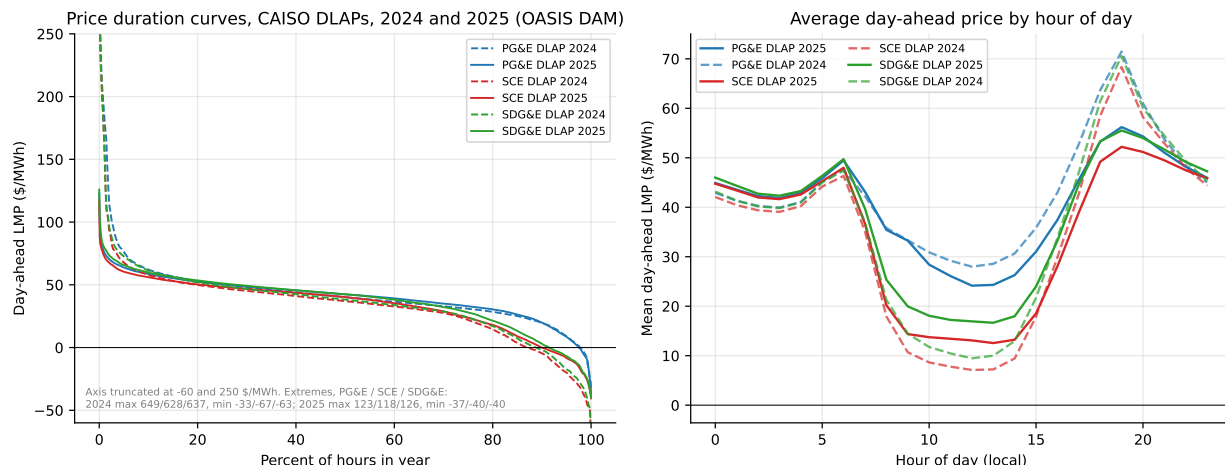


Figure 2.5: Price duration curves (left) and average price by hour of day (right), 2024 and 2025; one colour per DLAP, dashed 2024 and solid 2025. The duration axis is truncated at -60 and 250 $\$/\text{MWh}$ and the extremes are printed in the panel (2024 maxima of 628 to 649 $\$/\text{MWh}$). Source: CAISO OASIS.

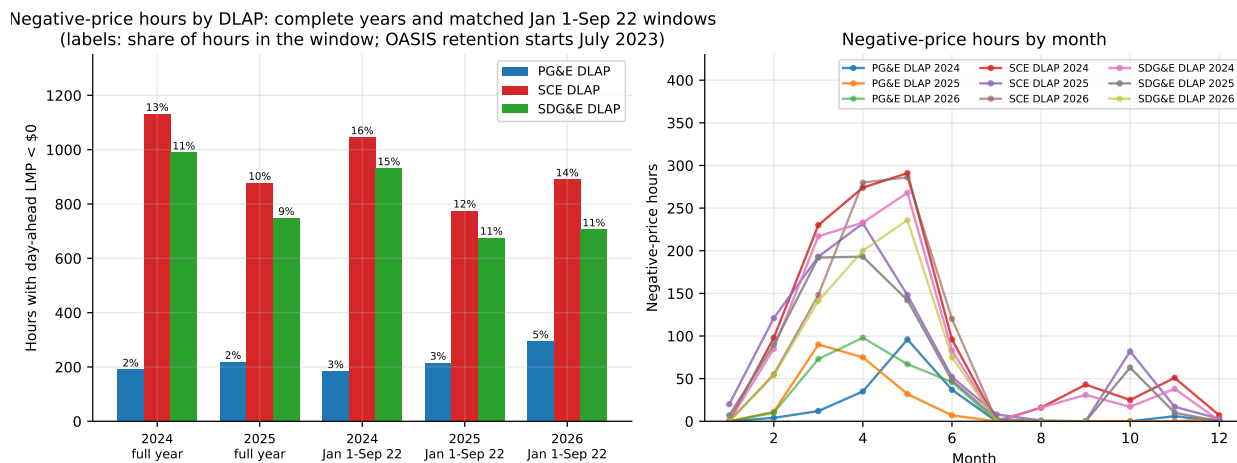


Figure 2.6: Negative-price hours at the three DLAPs. Left: the complete years 2024 and 2025 and matched January 1 to September 22 windows for 2024, 2025 and 2026, with the share of hours in each window; 2023 (July to December only) is omitted from the comparison. Right: by month. Source: CAISO OASIS.

Negative day-ahead prices are now routine in Southern California (Table 2.3, Figures 2.5 and 2.6). The SCE aggregation point cleared below zero in 1,131 hours of 2024 and 877 hours of 2025, and in 890 hours of 2026 through September 22; SDG&E's point saw 989 and 748 hours; PG&E's 190 and 218. On the matched January 1 to September 22 window the SCE point had 1,045 negative hours in 2024, 775 in 2025 and 890 in 2026, 16, 12 and 14 percent of the hours in the window, so 2026 sits between the two complete years rather than above them (`ch1_negative_price_hours_windows.csv`). Between 72 and 94 percent of those hours fall between 10:00 and 16:00, concentrated in February through June, and the average price by hour of day drops to $\$7$ – $\$30$ per MWh at midday against $\$50$ – $\$70$ in the evening peak. The daily ICE

block prices show almost no negative days, which is why the hourly series is required. This is the California-specific fact that Chapter 5 uses: a data center able to place consumption in those hours faces a zero or negative marginal energy price, and one that cannot adds to the evening hours where prices and net load are highest.

2.3 Hourly carbon intensity

CAISO’s Today’s Outlook reports five-minute CO₂ by source, with imports carried at CAISO’s default factor and export emissions netted against them [California Independent System Operator, 2026d]. The indicator used here is that accounting total divided by CAISO demand, energy-weighted per year: an accounting rate for the supply serving CAISO demand, not a measured consumption footprint. Three properties of the data are made explicit. First, because exports are netted, net CO₂ is negative in export-heavy hours (374 in 2024, 425 in 2025); the headline series keeps those hours at their negative values, and a variant that floors them at zero is shown as a sensitivity (it raises 2025 from 186.3 to 187.7 g/kWh). Second, CAISO publishes a fixed grid of 288 five-minute rows for every calendar day, including the two daylight-saving transition days, so the repeated autumn hour and the skipped spring hour cannot be separated; an hour is counted as valid when at least six of its twelve intervals exist for both CO₂ and demand, hours with no source data are left missing rather than counted as zero, and coverage is 99.8 percent of calendar hours or better in every year. Third, EIA’s warning that CAISO’s natural gas data collection changed in December 2023 [U.S. Energy Information Administration, 2025b] applies to the EIA-930 gas series, not to CAISO’s own accounting: the emission factor implied by CAISO’s gas CO₂ over CAISO’s gas generation stays between 432 and 452 kg/MWh in every month from June 2023 to June 2024 and between 437 and 458 in every year, and CAISO’s gas share of demand moves from 48.8 to 47.8 to 47.9 percent across November, December and January. The EPA eGRID output rate for the CAMX subregion is drawn as a reference with a different boundary: a generation-based rate for the whole subregion with no import factor [U.S. Environmental Protection Agency, 2025].

Table 2.4: CO₂ accounting intensity of CAISO demand: CAISO’s CO₂ accounting (imports net of exports) divided by CAISO demand, energy-weighted, with the floored variant, the number of negative hours and valid hours as a share of the calendar. eGRID CAMX is a generation output rate with a different boundary. Source: CAISO Today’s Outlook, eGRID 2023.

Year	Accounting (g/kWh)	Floored (g/kWh)	Negative hours	P5 (g/kWh)	P95 (g/kWh)	CO ₂ (Mt)	Imports share (%)	Gas share (%)	Valid hours (%)
2019	256.7	256.8	8	109	361	55.9	42	53	100.0
2020	277.4	277.4	0	132	368	60.4	42	54	100.0
2021	274.5	274.5	0	125	361	59.7	37	59	100.0
2022	264.4	264.4	13	108	355	58.1	34	63	100.0
2023	241.9	242.0	19	85	344	51.1	21	75	100.0
2024	199.4	200.7	374	7	322	42.4	27	68	99.8
2025	186.3	187.7	425	1	319	38.8	35	61	100.0
eGRID 2023 CAMX generation output rate	198.1	—	—	—	—	47.3	—	—	—

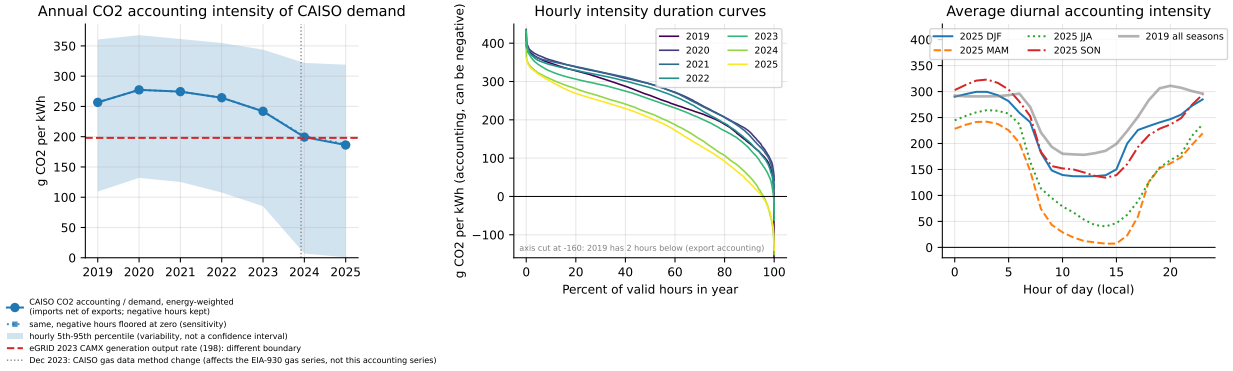


Figure 2.7: Annual, duration-curve and diurnal views of the CAISO CO₂ accounting intensity. Left: headline series with negative hours kept, the floored sensitivity, the hourly 5th to 95th percentile band (variability, not a confidence interval), the eGRID reference and the December 2023 EIA-930 gas reporting change. Middle: hourly duration curves, axis cut at -160 g/kWh. Right: 2025 seasonal diurnal means against 2019. Source: CAISO Today’s Outlook, eGRID 2023.

Table 2.5: CAISO battery fleet energy and peak power from the five-minute fuel mix.

Year	Discharge (TWh)	Charge (TWh)	Peak discharge (GW)	Peak charge (GW)
2019	0.05	0.08	0.1	0.1
2020	0.06	0.08	0.2	0.2
2021	0.83	0.92	1.4	1.2
2022	2.73	1.70	3.0	2.1
2023	4.34	4.45	4.8	4.3
2024	8.18	9.45	7.8	7.2
2025	12.29	14.21	10.7	9.3

The accounting intensity fell from 257 g/kWh in 2019 to 186 g/kWh in 2025 (Table 2.4, Figure 2.7); the floored variant gives 257 and 188. The hourly spread widened: in 2025 the fifth percentile was 1 g/kWh and the ninety-fifth 319 g/kWh, and the average midday hour was about 10 g/kWh in spring and 46 g/kWh in summer against roughly 240 to 315 g/kWh at night. eGRID’s 2023 CAMX generation rate of 198 g/kWh is lower than the 2023 accounting figure of 242 g/kWh because it excludes the import factor and covers the whole subregion; it is a single-year reference, not a trend. The decline is consistent with the solar and battery additions of Table 2.1, but the series remains an accounting construct: a validated consumption-emissions series would need CAISO’s greenhouse gas tracking reports as an independent check, which is deferred to the supply chapter. For a flat load the annual factor is the relevant one; for a flexible load the diurnal ratio, about five to one in summer and far larger in spring, is the relevant number, and an average intensity is not a marginal emissions factor for load shifting.

2.4 How much of the state’s power do data centers use

No agency meters California’s data center consumption, and the four items in Table 2.6 and Figure 2.8 are not four comparable estimates of one quantity. They are grouped by what they are: one statewide historical estimate, one assumed conversion of a published peak, one illustrative count-based sensitivity and one utility subset. Shares are computed against EIA-861 California retail

sales for the year that matches each item, 239.5 TWh in 2023 and 245.7 TWh in 2024 (Parts A, C and D, which avoids counting delivery-only sales twice) [U.S. Energy Information Administration, 2025a].

Table 2.6: Evidence and assumptions for existing California data center electricity use, by type. Shares use EIA-861 retail sales of the year in parentheses.

Group	Method	Data year	TWh per year	Share of retail sales, % (year)
A. Statewide estimate	EPRI 2024 state table, 2023 consumption	2023	9.3	3.9 (2023)
B. Assumed conversion	CEC 1,000 MW existing peak (Dec 2025) x assumed annual load factor 0.80-0.95	2025	7.0–8.3	2.9–3.4 (2024)
C. Illustrative sensitivity	Count-based: 321 Kollar-Grady facility points x assumed energy per facility	2023	6.8–13.6	2.8–5.7 (2023)
D. Utility subset	Silicon Valley Power only: 53-60% of SVP 2023 retail sales	2023	2.4–2.7	1.0–1.1 (2023)

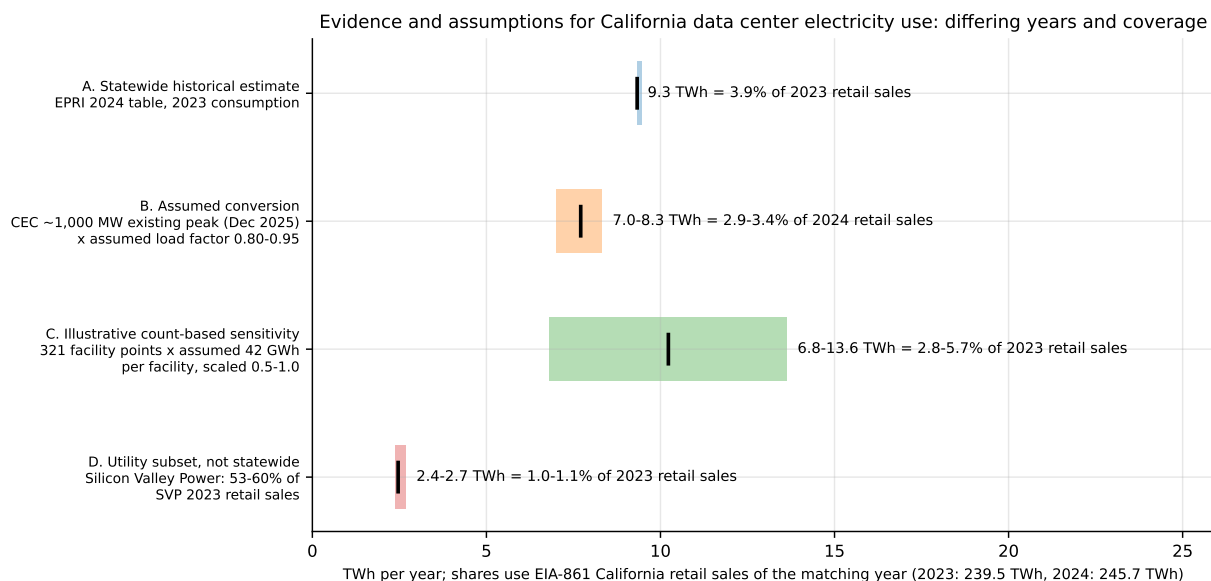


Figure 2.8: Existing data center electricity use in California: one statewide estimate (A), one assumed conversion (B), one illustrative sensitivity (C) and one utility subset (D), with matching-year retail-sales shares. No metered statewide total exists.

- A. Statewide estimate.** EPRI's 2024 study estimates California data centers consumed 9,331,619 MWh in 2023, 3.70 percent of state consumption on EPRI's denominator, the third-largest state total after Virginia and Texas [Electric Power Research Institute, 2024, pp. 13, 28]; against EIA-861 2023 retail sales it is 3.9 percent. This is the only statewide historical estimate in the set.
- B. Assumed conversion.** The CEC's methodology memo states that "as of December 2025, approximately 1,000 megawatts (MW) of existing data center peak demand operates in California" [California Energy Commission, 2026d, pp. 2, 18]. Converting a peak into energy needs an annual load factor that no source reports. The CEC's daily load-factor profiles for existing facilities range from 75 to 100 percent of the daily maximum [California Energy

Commission, 2025], and an assumed annual factor of 0.80 to 0.95 turns the 1,000 MW into 7.0 to 8.3 TWh, 2.9 to 3.4 percent of 2024 retail sales. It is a conversion scenario, not a measured total.

3. **C. Illustrative sensitivity.** The Kollar and Grady inventory places 321 of its 2,484 facility points inside California, 102 of them within 12 km of Santa Clara [Kollar and Grady, 2025], but carries no per-facility power, and the Epoch AI hub snapshot contains no California observation [Epoch AI, 2026a], which is a limit of its coverage rather than a count of zero. The sensitivity multiplies the count by an assumed annual energy per facility taken from the Silicon Valley Power cluster: 42 GWh, or 55 percent of SVP’s 2023 retail sales spread over its 58 data centers, an average load of 4.8 MW, scaled by 0.5, 0.75 and 1.0 because the statewide inventory includes many small facilities. The result, 6.8 to 13.6 TWh or 2.8 to 5.7 percent, uses no peak-demand or utilization factor. It is an assumption set whose representativeness is untested, shown to bound the statewide figure rather than to estimate it.
4. **D. Utility subset.** The municipal utility of Santa Clara reports 58 data centers, the largest cluster on the West Coast, taking 55 percent of its power use [Silicon Valley Power, 2026a]; its web page states 53 percent [Silicon Valley Power, 2026b], and city officials cited about 60 percent in 2025 [San José Spotlight, 2025]. Applied to SVP’s 2023 retail sales of 4.48 TWh [Silicon Valley Power, 2024] (its purchased and generated supply was 4.59 TWh, which is what a peak of 669.2 MW times a 78.3 percent load factor reconstructs), the cluster used 2.4 to 2.7 TWh, about one percent of statewide 2023 sales; the percentages come from different dates than the sales figure. SVP also reports that its best-performing data center customers reach about 67 percent utilization of requested capacity and that observed utilization was about 64 percent in 2023 [Silicon Valley Power, 2025], the empirical origin of the CEC’s 67 percent factor; that ratio relates operating peak to requested capacity and is not an annual load factor.

The statewide estimate and the assumed conversion agree at 7 to 9 TWh, roughly 3 to 4 percent of retail sales, and the sensitivity admits 3 to 6 percent. One municipal utility hosts about a quarter to a third of that. The defensible statement for the rest of this study is that existing data centers used about 3 to 4 percent of California’s electricity in 2023 to 2025, that the load is geographically concentrated, and that no metered statewide total exists. Against roughly 1 GW of existing peak, the CEC’s December 2025 energization requests located in California total 20,677 MW, or 23,277 MW including the 2,600 MW that VEA reports for Nevada [California Energy Commission, 2026d], a ratio of about 21 that is the California analogue of Finland’s 27x [Manner, 2026] and the subject of Chapter 3.

2.5 Institutional context

The January 2026 forecast decision. The CEC adopted the California Energy Demand 2025–2045 forecast set on January 21, 2026 [California Energy Commission, 2026c]. The set includes three data center scenarios built from the energization tiers. What the joint agencies then chose to do with those scenarios is the institutional fact that matters. The Single Forecast Set Agreement between the CEC, the CPUC and the CAISO, signed in January 2026 and updated on April 15, 2026, directs that CPUC integrated resource planning and CAISO bulk-system transmission studies for the 2026–2027 cycle “will continue to use the 2024 IEPR planning forecast,” that the 2027–2028 cycle will use an IRP portfolio based on the same 2024 forecast, and that system resource

adequacy requirements for 2027 “will use the 2025 IEPR planning forecast without known loads” [California Energy Commission, California Public Utilities Commission and California Independent System Operator, 2026, pp. 7–8]. Only local transmission reliability studies use the 2025 IEPR local reliability forecast with known loads. The agencies give two reasons: the approach “manages volatility and promotes stability in electric system resource and infrastructure planning currently underway,” and it “allows CEC staff additional time to collect and review historical data, observe the impacts of known loads, data centers, and recent federal policy changes, and refine forecast approaches as needed” [California Energy Commission, California Public Utilities Commission and California Independent System Operator, 2026, p. 8]. In plain terms, the state now tracks data center energization requests in tiers, converts them into forecast scenarios, and then holds the resulting increment out of the forecast that drives generation procurement and bulk transmission for at least two planning cycles. The 2024 IEPR data center mid case, built from the smaller December 2024 tiers, remains the number in use.

Utility pipeline statements. PG&E told the CEC in February 2025 that it had “a large pipeline of applications to interconnect new data centers” with about 1.5 GW in final engineering or construction and more than half expected within four years [Nelli, 2025]. In July 2025 it announced a 10 GW pipeline over ten years, of which 17 projects totaling about 1.5 GW were in final engineering for operation between 2026 and 2030, and estimated that each gigawatt of new data center demand could lower other customers’ bills by one to two percent [Pacific Gas and Electric Company, 2025]. Its second-quarter 2026 earnings release describes “an overall pipeline of over 12 gigawatts” [PG&E Corporation, 2026]. PG&E has, however, filed a request for confidentiality over its project-level data center data in the IEPR docket every month since August 2025 [Pacific Gas and Electric Company, 2026a], so its pipeline is public only in aggregate. Southern California Edison took the opposite path. It reported 80 MW of existing data center demand in February 2025 with 200 MW of near-term and 400 MW of mid-term growth [Dean, 2025], and it files a public project-level database. The January 2026 vintage lists 115 entries with 8,298 MW of requested peak: 76 MW under signed agreements, 3,314 MW in applications, 1,773 MW in inquiries, and 3,137 MW already canceled [Southern California Edison, 2026]. The CEC’s own summer 2025 tally put SCE’s requests at 5,828 MW and PG&E’s at 11,668 MW [California Energy Commission, 2025]. The canceled share in SCE’s file, 38 percent of everything ever requested, is the first direct California evidence on attrition and is used in Chapter 5.

CPUC large-load proceedings. The proposal could not confirm a CPUC large-load tariff docket; the record now can. On April 9, 2026 the CPUC opened Rulemaking 26-04-009 on Advanced Electric Rate Design, which states that “in coordination with the Senate Bill 57 assessment, this proceeding will consider rate design issues for data centers and other large load customers” [California Public Utilities Commission, 2026, p. 1]. Senate Bill 57 (Padilla), chaptered on October 11, 2025, requires the CPUC to assess whether costs of serving new data center loads shift to other customers and to report to the Legislature by January 1, 2027 [California Legislature, 2025b]. Senate Bill 886 (Padilla), the California Technology Innovation and Ratepayer Protection Act, was chaptered on September 21, 2026 and requires the CPUC, on or before January 1, 2028, to establish new tariffs or update electric rules for the interconnection and service of participating large customers, and to ensure those tariffs “prevent the creation of stranded costs for, or cost shifts to, nonparticipating customers” [California Legislature, 2026, KQED, 2026]. Assembly Bill 222, which would have required data center energy reporting and efficiency metrics, was held in the Senate Appropriations Committee on August 29, 2025 [California Legislature, 2025a]. On the

interconnection side, the CPUC approved an interim PG&E Electric Rule 30 on July 24, 2025 to streamline transmission-level service for large customers that fund the required infrastructure up front [California Public Utilities Commission, 2025]. The sequence is consistent: the state has moved from studying the cost question (SB 57) to mandating a tariff answer (SB 886) within a year, while the forecast that drives procurement still excludes the load the tariff is meant to price.

Chapter 3

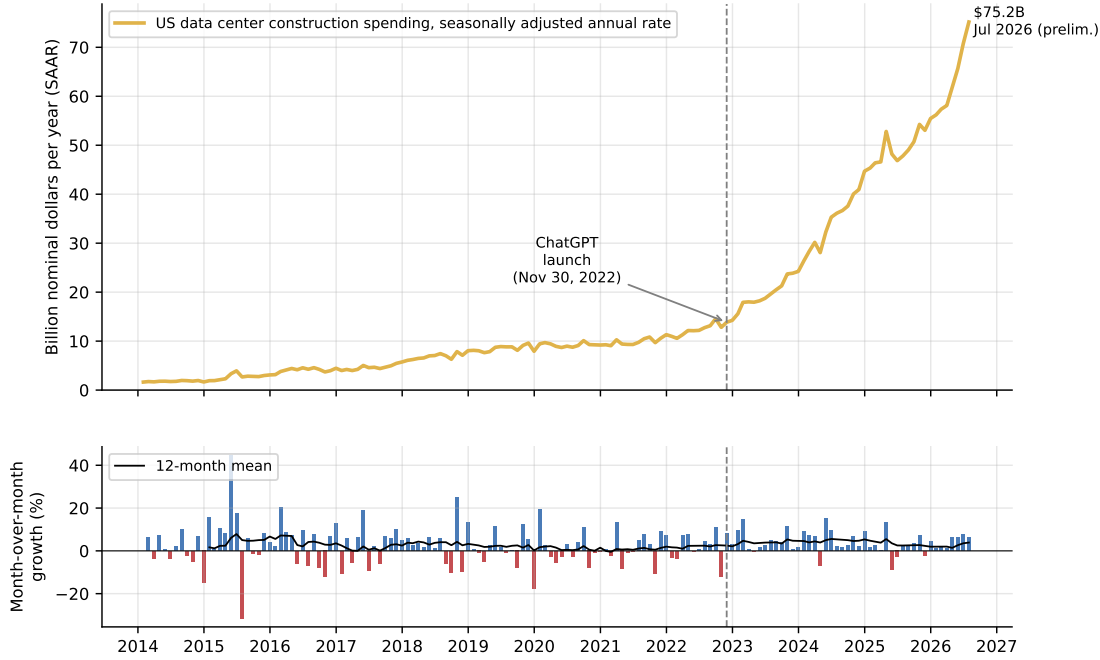
Data Center Growth in California

This chapter answers RQ1: how California’s planned data center capacity varies by stage of development, in absolute terms and relative to the scale of the existing grid. It first reproduces the national construction-spending series that frames public discussion, tests it for structural breaks, and then asks whether California’s own indicators show the same acceleration. It ends with every vintage of the CEC’s energization tiers and the RQ1 table. All code is in `src/ch2_growth.py` and `notebooks/02_data_center_growth.ipynb`.

3.1 The national construction boom, reproduced

The Census Bureau’s Value of Construction Put in Place survey has carried a “data center” line under private nonresidential construction since January 2014 [U.S. Census Bureau, 2026]. Figure 3.1 reproduces the widely circulated chart from the raw seasonally adjusted workbook and adds month-over-month growth. Spending at a seasonally adjusted annual rate rose from \$1.6 billion in January 2014 to \$13.9 billion in November 2022 and \$75.2 billion in July 2026 (preliminary). Data centers were half a percent of private nonresidential construction in 2014, two percent when ChatGPT launched, and ten percent in July 2026. Month-over-month growth averaged 4.1 percent from 2023 through mid-2026 against 1.2 percent in 2019 through 2021. No comparable state-level series exists: the Census publishes this category for the nation only, and the California picture must be built from proxies (Section 3.3).

US private construction spending, data center category (nominal dollars, seasonally adjusted annual rate), Jan 2014 to Jul 2026 (Census C30)



Source: U.S. Census Bureau, Value of Construction Put in Place (privsatime.xlsx), accessed 2026-09-21; p = preliminary

Figure 3.1: US private data center construction spending, seasonally adjusted annual rate, with month-over-month growth. Source: Census C30, July 2026 preliminary.

3.2 Structural breaks and forecasts

The analysis works on the log of the seasonally adjusted series, so that a linear trend is a constant compound growth rate. Three questions are asked in turn: does a break at the known date change the trend (Chow test, Chow, 1960); where does the data itself place breaks when none is imposed (Bai-Perron, Bai and Perron, 1998, 2003); and what does a time-series model project. Growth rates carry Newey-West standard errors. The Bai-Perron search uses 15 percent trimming (a minimum segment of 23 months) and allows up to five breaks, the most the trimming permits, so that the choice of m is never censored by the search. The number of breaks is chosen three ways: by BIC, by the LWZ criterion, and by the sequential $\text{supF}(l+1|l)$ procedure, in which each additional break is tested against the l -break model with a moving-block residual bootstrap p -value. $\text{supF}(0|1)$ gets the same bootstrap under the no-break null, and the single-break date a residual-bootstrap 90 percent interval. The `ruptures` dynamic-programming search [Truong et al., 2020] is run as an independent check.

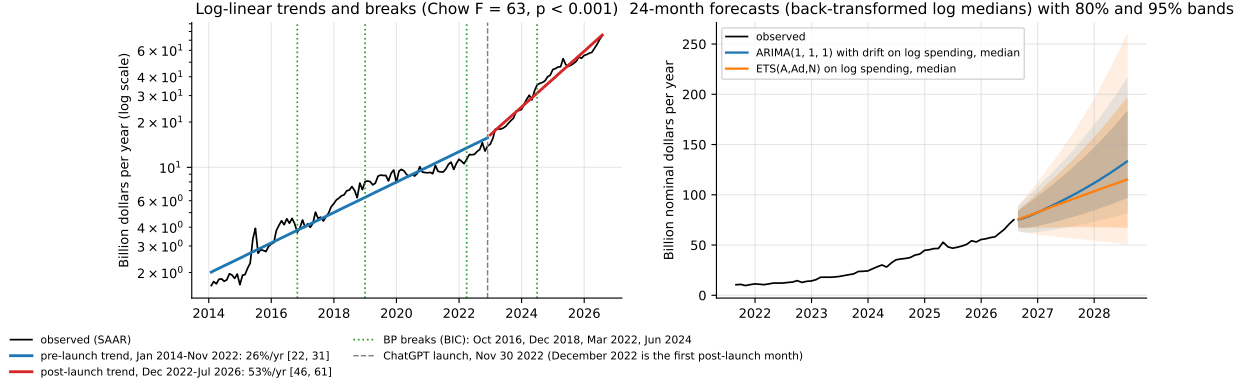


Figure 3.2: Left: log-linear trends before and after December 2022 with Bai-Perron break dates. Right: 24-month ARIMA and ETS forecasts with 80 and 95 percent prediction bands.

Table 3.1: Bai-Perron global search on the log Census series: break dates, sum of squared residuals, information criteria, $\text{supF}(0|m)$, and the sequential $\text{supF}(m|m-1)$ with its bootstrap p -value, for $m = 0$ to 5 breaks.

Breaks m	Break dates (first month of new regime)	SSR	BIC	LWZ	$\text{supF}(0 m)$	$\text{supF}(m m-1)$	boot. p
0	—	5.967	-3.165	-3.101	—	—	—
1	2020-11	2.205	-4.061	-3.900	125.4	125.4	0.020
2	2018-12, 2022-03	1.375	-4.433	-4.176	121.1	36.7	0.085
3	2016-10, 2018-12, 2022-03	1.091	-4.565	-4.210	106.5	40.6	0.020
4	2016-10, 2018-12, 2022-03, 2024-06	0.880	-4.680	-4.228	101.9	40.6	0.005
5	2016-02, 2018-01, 2020-04, 2022-05, 2024-06	0.832	-4.636	-4.086	85.7	—	—

Table 3.2: Compound growth by segment of the BIC-selected partition (Newey-West 95 percent intervals).

Segment (BIC partition)	Months	Compound growth (%/yr)	95% interval
2014-01 to 2016-09	33	53	[45, 61]
2016-10 to 2018-11	26	42	[37, 47]
2018-12 to 2022-02	39	9	[7, 10]
2022-03 to 2024-05	27	61	[56, 67]
2024-06 to 2026-07	26	36	[31, 41]

The known-date test is unambiguous. A break in intercept and slope at December 2022, the first full month after the November 30 launch, gives a Chow F of 63 ($p < 0.001$) and the HAC-robust Wald test agrees ($p < 0.001$). Compound growth was 26 percent a year (95 percent interval 22 to 31) through November 2022 and 53 percent a year (46 to 61) from December 2022. Neither the date annotation nor the test establishes that the launch caused the acceleration; they establish that the trend changed then. Letting the data locate the breaks changes the story in two respects (Tables 3.1 and 3.2). BIC and LWZ both choose four breaks, in October 2016, December 2018, March 2022 and June 2024. The current expansion therefore began eight months before ChatGPT, in the hyperscaler capital-spending ramp of early 2022, and it has already slowed once: the segment

from March 2022 to May 2024 grew at 61 percent a year (56 to 67) and the segment from June 2024 to July 2026 at 36 percent a year (31 to 41), still four times the 9 percent of 2019 to early 2022. The sequential procedure is more conservative. $\text{supF}(1|0)$ of 125 rejects the no-break model ($p = 0.02$), but $\text{supF}(2|1)$ of 37 has a bootstrap p -value of 0.085, so at the 5 percent level the procedure stops at one break, in November 2020, even though the third and fourth breaks are each significant once the earlier ones are in place ($p = 0.02$ and $p = 0.005$). **ruptures** reproduces all four BIC dates within one month. The forecasts in Table 3.3 continue the recent trend: an ARIMA(1,1,1) with drift and a damped-trend ETS, both fitted to log spending, put July 2027 near \$95 to \$100 billion and July 2028 near \$115 to \$135 billion, with 95 percent bands from roughly \$50 to \$260 billion by then. The point forecasts are back-transformed log medians, not arithmetic means, and the series is national nominal dollars, so the forecasts say nothing about California megawatts. A rolling-origin backtest over the last 24 months (Table 3.4) shows why they are exploratory: at a twelve-month horizon the mean absolute percentage error is 14 percent for the ARIMA and 19 percent for the ETS against 13 percent for a log random walk with drift, every model ran below the outcomes on average, and the 95 percent bands covered 92 percent of outcomes for the ARIMA and 79 percent for the ETS. The bands, not the point forecasts, are the finding: a series whose growth rate has shifted four times in twelve years, and which a drift benchmark forecasts as well as either model, is not something a planner should extrapolate.

Table 3.3: Forecasts of US data center construction spending, 6, 12 and 24 months ahead: back-transformed log medians with 80 and 95 percent bands, nominal dollars.

Month	ARIMA(1, 1, 1) with drift			ETS(A,Ad,N)		
	Mean	80% band	95% band	Mean	80% band	95% band
Jan 2027	84.5	[70.7, 101.0]	[64.3, 111.0]	84.4	[68.0, 104.7]	[60.6, 117.4]
Jul 2027	98.3	[78.0, 124.0]	[68.9, 140.2]	94.8	[68.4, 131.3]	[57.6, 156.0]
Jul 2028	133.2	[97.4, 182.1]	[82.5, 214.9]	115.1	[67.8, 195.6]	[51.2, 258.9]

Billion dollars per year, seasonally adjusted annual rate.

Table 3.4: Rolling-origin backtest over the last 24 months: mean absolute percentage error on the level scale, root-mean-square log error, mean log error (bias) and band coverage, by horizon, for the two models and two benchmarks.

Model	1 month ahead				6 months ahead				12 months ahead			
	MAPE (%)	RMSE (log)	bias (log)	cover. 80/95	MAPE (%)	RMSE (log)	bias (log)	cover. 80/95	MAPE (%)	RMSE (log)	bias (log)	cover. 80/95
ARIMA (log, drift)	4.6	0.06	-0.02	96 / 100	11.6	0.14	-0.08	75 / 100	14.0	0.20	-0.11	67 / 92
ETS(A,Ad,N) (log)	4.9	0.06	-0.03	96 / 100	13.9	0.18	-0.11	75 / 92	18.8	0.30	-0.20	67 / 79
drift benchmark (log random walk)	4.0	0.06	-0.01	- / -	10.2	0.12	-0.06	- / -	13.0	0.18	-0.10	- / -
naive benchmark (last value)	5.1	0.07	-0.04	- / -	18.2	0.23	-0.21	- / -	31.2	0.41	-0.39	- / -

3.3 California proxies

Three California indicators are subjected to the same battery, with the known break set to the first period after November 2022 (Figure 3.3, Table 3.5).

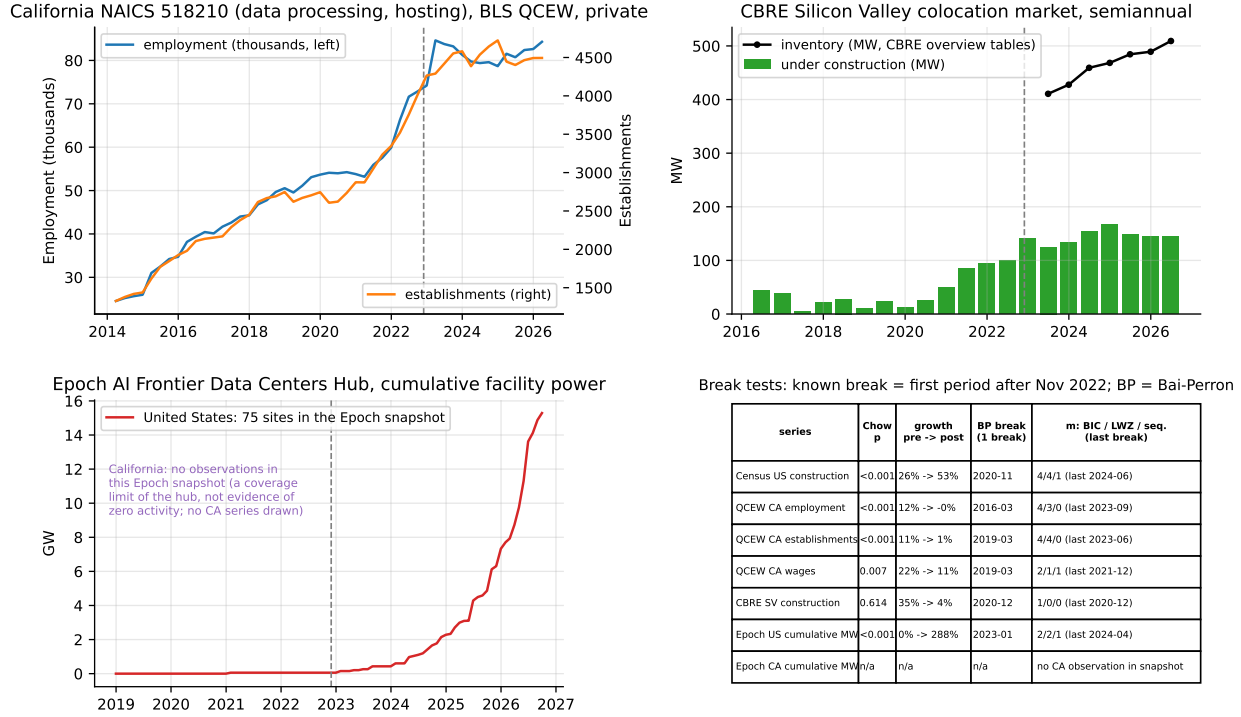


Figure 3.3: California proxies for data center growth: BLS QCEW NAICS 518210 labour indicators (top left), CBRE Silicon Valley construction and inventory (top right), Epoch AI frontier sites, whose snapshot has no California observation (bottom left), and the break-test summary (bottom right); the single-break column is the one-break model’s estimate and p-values below 0.001 are shown as such.

Table 3.5: Break tests on the Census series and the California proxies. Chow: known break at the first period after November 2022. BP: Bai-Perron single break with residual-bootstrap 90 percent interval, the number of breaks chosen by BIC, LWZ and the sequential procedure, the BIC-selected dates, and the bootstrap p -value of $\sup F(0|1)$.

Series	<i>n</i>	Chow <i>F</i>	<i>p</i>	HAC <i>p</i>	Growth before (%/yr)	Growth after (%/yr)	BP break (1)	90% interval	<i>m</i> : BIC/LWZ/seq.	BP breaks (BIC)	boot. <i>p</i>
Census data center construction (monthly, SAAR)	151	63.0	0.000	0.000	26 [22, 31]	53 [46, 61]	2020-11	2020-11 to 2020-11	4/4/1	2016-10, 2018-12, 2022-03, 2024-06	0.010
QCEW 518210 California employment (quarterly)	49	15.0	0.000	0.000	12 [9, 14]	-0 [-2, 2]	2016-03	2016-03 to 2021-12	4/3/0	2016-03, 2018-09, 2021-03, 2023-09	0.035
QCEW 518210 California establishments (quarterly)	49	11.0	0.000	0.000	11 [9, 13]	1 [-1, 2]	2019-03	2016-12 to 2022-09	4/4/0	2016-03, 2018-03, 2020-12, 2023-06	0.285
QCEW 518210 California wages (quarterly)	49	5.5	0.007	0.000	22 [19, 24]	11 [9, 13]	2019-03	2016-12 to 2019-06	2/1/1	2019-03, 2021-12	0.005
CBRE Silicon Valley MW under construction (semiannual)	21	0.5	0.614	0.000	35 [10, 66]	4 [-2, 10]	2020-12	2020-12 to 2020-12	1/0/0	2020-12	0.140
Epoch frontier sites, United States cumulative MW (monthly)	69	176.1	0.000	0.000	0 [-0, 0]	288 [250, 331]	2023-01	2023-01 to 2023-01	2/2/1	2023-01, 2024-04	0.005
Epoch frontier sites, California cumulative MW	0	no California observation in the Epoch snapshot (0 of 87 sites): a coverage limit of the h									

Employment and establishments. The Quarterly Census of Employment and Wages counts private establishments and jobs in NAICS 518210, computing infrastructure, data processing, hosting and related services, by quarter [U.S. Bureau of Labor Statistics, 2026]. These are labour indicators of an industry, not counts of buildings or megawatts, and the industry includes hosting and processing firms that do not operate data centers. California employment in the industry rose from 25,000 in 2014 Q1 to 74,000 by the end of 2022, then flattened: 84,000 in 2026 Q1 after a plateau through 2024 and 2025. The Chow test at 2023 Q1 rejects a common trend ($F = 15$, $p < 0.001$), but the break is a deceleration, from 12 percent a year to roughly zero. Establishments

show the same shape, 11 percent a year to 0.5 percent ($F = 11$). Data-located breaks in these smooth quarterly series are diffuse: BIC places four in each, the last in 2023 Q3 for employment and 2023 Q2 for establishments, but the single-break bootstrap is significant only for employment ($p = 0.035$ against 0.285 for establishments) and the sequential procedure keeps no break in either. Wages decelerated from 22 to 11 percent a year ($p = 0.007$) with breaks in 2019 Q1 and 2021 Q4 that reflect the industry’s earlier expansion. The county detail is in the notebook; Santa Clara, Los Angeles and San Francisco account for most of the state total.

MW under construction. CBRE’s semiannual market chapters give Silicon Valley colocation capacity under construction from H1 2016 and inventory from H1 2023 [CBRE Research, 2026]. Construction rose from 44 MW in H1 2016 to 142 MW by H2 2022 and has since ranged between 125 and 168 MW, with inventory creeping from 411 to 509 MW. Megawatts under construction are a stock whose change reflects completions as well as starts, and the series covers one colocation market, not California construction spending. The Chow test finds no break at H1 2023 ($p = 0.61$). BIC keeps a single break in H2 2020, when the pipeline stepped up from tens of megawatts to over a hundred, and growth has been 4 percent a year since; LWZ and the sequential procedure keep no break at all (bootstrap $p = 0.14$), which is what a plateau looks like in 21 semiannual points. Cushman & Wakefield’s tables are form-gated and JLL’s report gives a single-period inventory, so the digitized series rests on CBRE; Southern California is available only from H1 2024 and is reported in the notebook.

Frontier facilities. Epoch AI’s hub tracks 87 large AI data centers from satellite imagery and permits, 75 of them in the United States, with per-site power timelines [Epoch AI, 2026b]. Cumulative US power in these sites went from under 0.5 GW at the end of 2023 to 15.3 GW by September 2026, with a break in January 2023; the curve is a reconstruction from sampled sites, and its breaks can reflect the hub’s coverage and retrospective estimates as much as construction. None of the 87 sites in the snapshot is in California. That is a limit of the hub’s coverage, which Epoch itself describes as incomplete and uneven, not evidence that no frontier facility exists or is being built in California; no California series is drawn and no test is possible.

The three proxies agree on a finding that the national series hides: after 2022 California’s data center industry did not accelerate in jobs, establishments or built Silicon Valley colocation capacity, and the largest AI campuses in Epoch’s sample were sited elsewhere. The growth that California records is in requests to connect, which is the subject of the next section, and the divergence between requests and built capacity is exactly what the tiers are meant to measure.

3.4 Every vintage of the energization tiers

Table 3.6 and Figure 3.4 assemble every published vintage of the CEC’s data center energization requests, followed by the two utility-level series that carry the record past December 2025. The December 2024 data behind the 2024 IEPR Update are known only as PG&E’s and SCE’s agreements plus applications, 5,808 and 963 MW, because inquiries were not published [California Energy Commission, 2025]. The summer 2025 data behind the preliminary 2025 forecast are published as totals by utility, 21,756 MW across seven utilities; for PG&E and SCE the deck also gives agreements plus applications, 10,080 and 143 MW, so their inquiries were 1,588 and 5,685 MW [California Energy Commission, 2025]. The copy of that deck filed for the November 12, 2025 workshop labelled SCE at 2,492 MW and SVP at 1,382 MW; the published deck corrects these to 143 and 1,375 MW, and the corrected figures are used here. The December 2025 data behind the adopted forecast are

the first complete vintage: 5,086 MW signed, 9,587 MW in application and 8,604 MW in inquiry, 23,277 MW in all, with PG&E accounting for 14,747 MW and SCE for 4,624 MW by the memo’s tier rows [California Energy Commission, 2026d,b]. The memo’s footnote states that VEA’s 2,600 MW of applications are for locations in Nevada, inside the CAISO footprint but outside California, and the CEC excludes them from statewide consumption; the pipeline located in California is therefore 20,677 MW (5,086 signed, 6,987 in application, 8,604 in inquiry), and the summer 2025 total contains the same 2,600 MW, leaving 19,156 MW in California. The published marginal totals and the utility rows differ by 1 MW (23,277 against 23,278); both are kept and the marginal totals are used. The August 2026 workshop restated the December 2025 figures and showed June 2026 known-load data only as charts [Smith, 2026]. SCE’s public project database provides the only vintages with cancellations: between its August 2025 and January 2026 versions, inquiries fell from 5,934 to 1,772 MW as about 3,100 MW moved into the application stage, canceled requests rose from 709 to 3,137 MW, and the active total fell from 6,201 to 5,162 MW [Southern California Edison, 2026]. The growth in SCE’s total records between the two snapshots is cancellations, not pipeline; thirty-eight percent of everything SCE had ever logged was canceled within five months, the first direct California evidence on attrition. PG&E’s pipeline table from its second-quarter 2026 earnings, reproduced in its August 2026 presentation to the CEC, uses PG&E’s own stages and excludes inquiries: 5,090 MW in March 2026 and 12,710 MW in June 2026, of which 3,880 MW had a signed work performance agreement, 490 MW an interconnection construction agreement and 140 MW was in construction [Pacific Gas and Electric Company, 2026b]. The jump of almost 8,000 MW in one quarter came from the 2026 cluster study. The Public Advocates Office observes that PG&E had filed agreements covering only about 650 MW with the CPUC for approval [Public Advocates Office, California Public Utilities Commission, 2026], so even the most committed tier has layers of certainty inside it. The three source families of Figure 3.4, the CEC’s statewide vintages, SCE’s database and PG&E’s earnings pipeline, differ in utility coverage, stage definitions and dates, so their bar heights are not one growth series, and signed agreements are commitments to connect, not operating capacity.

Table 3.6: Vintages of the CEC energization tiers found in the record (MW of requested capacity). SCE rows are that utility’s public project database. PG&E rows use PG&E’s own pipeline stages from its Q2 2026 earnings, where WPA+ means a signed work performance agreement or a later stage, and exclude inquiries.

Vintage	Signed	Applications	Inquiries	Agr.+appl. only	Unsplit total	Total active (MW)
Dec 2024 (2024 IEPR Update)	0	0	0	6,771	0	6,771
Summer 2025 (2025 IEPR preliminary)	0	0	7,273	10,223	4,260	21,756
Dec 2025 (2025 IEPR final)	5,086	9,587	8,605	0	0	23,278
Aug 2026 workshop (restates Dec 2025)	0	0	0	0	23,278	23,278
SCE database Aug 29 2025 (SCE only)	51	216	5,934	–	canceled 709	6,201
SCE database Jan 29 2026 (SCE only)	76	3,313	1,772	–	canceled 3,137	5,162
PG&E earnings pipeline Jun 2026 (PG&E only, PG&E stages)	WPA+ 4,510	8,200	n/a	12,710	–	12,710
PG&E earnings pipeline Mar 2026 (PG&E only, PG&E stages)	WPA+ 3,390	1,700	n/a	5,090	–	5,090

Totals are as reported by the utilities and include VEA’s 2,600 MW (summer 2025) and 2,600 MW (December 2025) of requests located in Nevada; California-only totals are 19,156 and 20,678 MW by utility rows (the CEC’s marginal totals are 1 MW lower: 23,277 and 20,677 MW). SCE active totals exclude canceled requests.

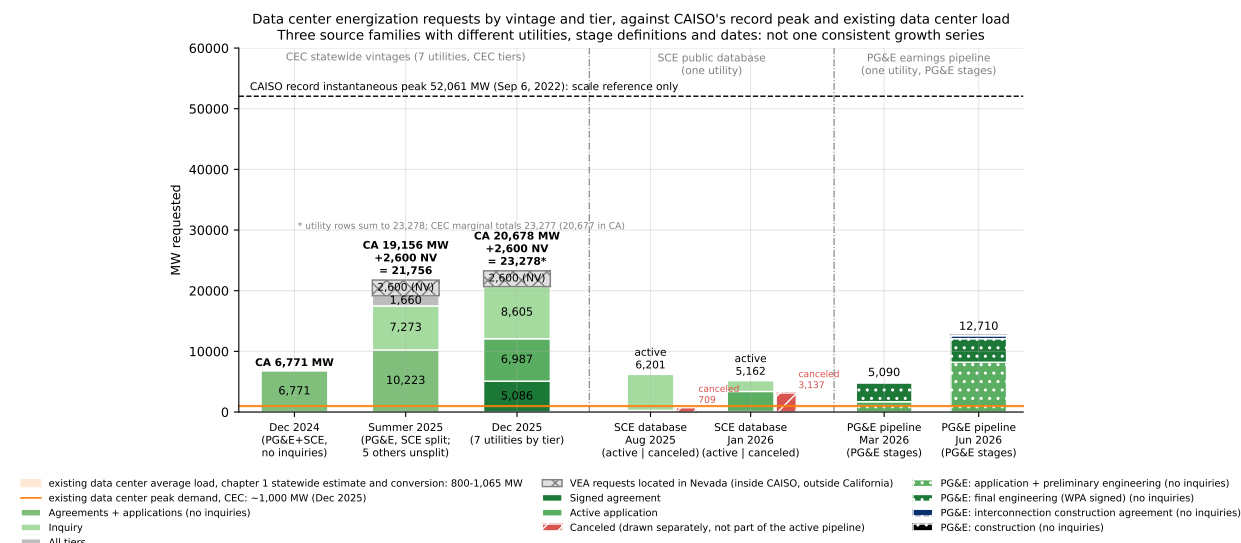


Figure 3.4: Energization requests by vintage and tier against CAISO’s record peak and the existing data center average load from Chapter 2. VEA’s requests in Nevada are hatched and California-only totals are labelled; SCE’s canceled requests are drawn beside, not on top of, its active pipeline; the three source families are divided because they differ in utilities, stages and dates.

The sums and ratios are stated explicitly. The three December 2025 tiers total 23,277 MW as reported and 20,677 MW located in California. Against CAISO’s record instantaneous peak of 52,061 MW on September 6, 2022 [California Independent System Operator, 2026b], the California pipeline is 40 percent (40 percent against the hourly EIA-930 peak of 51,104 MW, and 45 percent for the reported total including Nevada). This is a scale comparison, not a coincident-peak ratio and not a measure of usable headroom. Against the roughly 1,000 MW of existing data center peak demand that the CEC reports for December 2025, it is 21 times; against the 800 to 1,065 MW of existing average load implied by the chapter 1 statewide estimate and assumed conversion, 19 to 26 times. Finland’s ratio of registered to installed capacity was 27 [Manner, 2026]; California’s is of the same order.

3.5 RQ1: planned capacity by stage

Table 3.7: RQ1: California’s planned data center capacity by stage (December 2025, requests located in California) relative to the grid, to existing data center load, and to the adopted forecast’s growth to 2030; VEA’s Nevada requests and the CEC’s reported total are memo rows.

Stage (Dec 2025)	MW	% of CAISO record peak	% of 2025 peak	× existing DC peak	× existing DC avg. load	% of IEPB peak growth	% of IEPB growth at 67% util.
Signed agreements	5,086	9.8	11.6	5.1x	4.8-6.4x	127	85
Active applications, California	6,987	13.4	15.9	7.0x	6.6-8.7x	174	116
Inquiries	8,604	16.5	19.6	8.6x	8.1-10.8x	214	143
Total, three tiers, California only	20,677	39.7	47.1	20.7x	19.4-25.8x	514	345
Memo: VEA applications located in Nevada	2,600	5.0	5.9	2.6x	2.4-3.2x	65	43
Memo: CEC statewide total including VEA	23,277	44.7	53.1	23.3x	21.9-29.1x	579	388

Denominators: CAISO record peak 52,061 MW (Sept. 6, 2022); 2025 hourly peak 43,860 MW; existing data center peak ≈1,000 MW (CEC); existing average load 800–1,065 MW (chapter 1 range); IEPB planning-scenario CAISO managed net peak growth 2025–2030 4,019 MW.

Table 3.7 answers RQ1. The denominators come from the sources of the preceding chapters: the CAISO record and 2025 peaks, the CEC’s existing-load figure and the chapter 1 range, and the adopted 2025 IEPR planning forecast, whose hourly CAISO forecast gives a managed net peak of 46,479 MW in 2025 and 50,498 MW in 2030, a growth of 4,019 MW, and whose statewide energy to serve load grows from 263 to 298 TWh [California Energy Commission, 2026e]. Signed agreements alone, 5,086 MW, exceed the forecast’s entire five-year peak growth (127 percent of it); the three California tiers together are 5.1 times that growth (5.8 times including the Nevada requests), 3.4 times it even at the CEC’s 67 percent utilization factor, and their energy equivalent at that utilization and a 0.88 load factor, about 107 TWh a year, is 3.1 times the forecast’s statewide energy growth to 2030. Requests are not simultaneous demand: signed agreements are not operating capacity, and the forecast’s confidence and utilization factors exist to discount exactly this. The forecast is not blind to data centers: its planning scenario carries a data center contribution at the CAISO peak that rises from 96 MW in 2025 to 1,743 MW in 2030 (4,377 MW in the local-reliability scenario), and data-center-only deliveries of 12.1 TWh in 2030, roughly a third of projected statewide energy growth. What the forecast does is discount the requests, by the confidence levels and utilization of Chapter 2, to about a tenth of their face value at the peak by 2030. Whether that discount is right is the question the remaining chapters take up with the supply side, the attrition evidence and the adequacy framing.

Chapter 4

Energy Production Growth in California

This chapter builds the supply side of RQ2, which the proposal left thin. It records what California’s generating fleet produced and imported from 2010 to 2025, what capacity it holds and is scheduled to lose, and, from the state’s own planning record, how much of what is planned actually gets built and how late. The last section applies that empirical attrition to the current planned list to give probability-weighted supply for 2030 in three cases, symmetric with the confidence weighting the CEC applies to data center requests. Code is in `src/ch3_supply.py` and `notebooks/03_supply_growth.ipynb`; the pipeline parses sixteen EIA-923 and sixteen EIA-860 annual files and eighteen EIA-860M monthly files directly from the raw archives.

4.1 Generation by resource and imports, 2010–2025

In-state generation comes from the EIA-923 plant-level schedule, with each California plant’s reported fuel and prime mover classified into twelve resource classes [U.S. Energy Information Administration, 2026c]. The EIA-923 totals match the CEC’s in-state totals to within one percent in every year for which both exist. Net imports come from the CEC’s total-system series, which is statewide and includes the utilities outside CAISO [California Energy Commission, 2026a]; the CEC has not yet published 2025 imports, and its 2021 page repeats the 2020 table, so the multi-year table is used for the import series and the CAISO-only net interchange from Chapter 2 is shown as a footprint-limited proxy for 2019 to 2025.

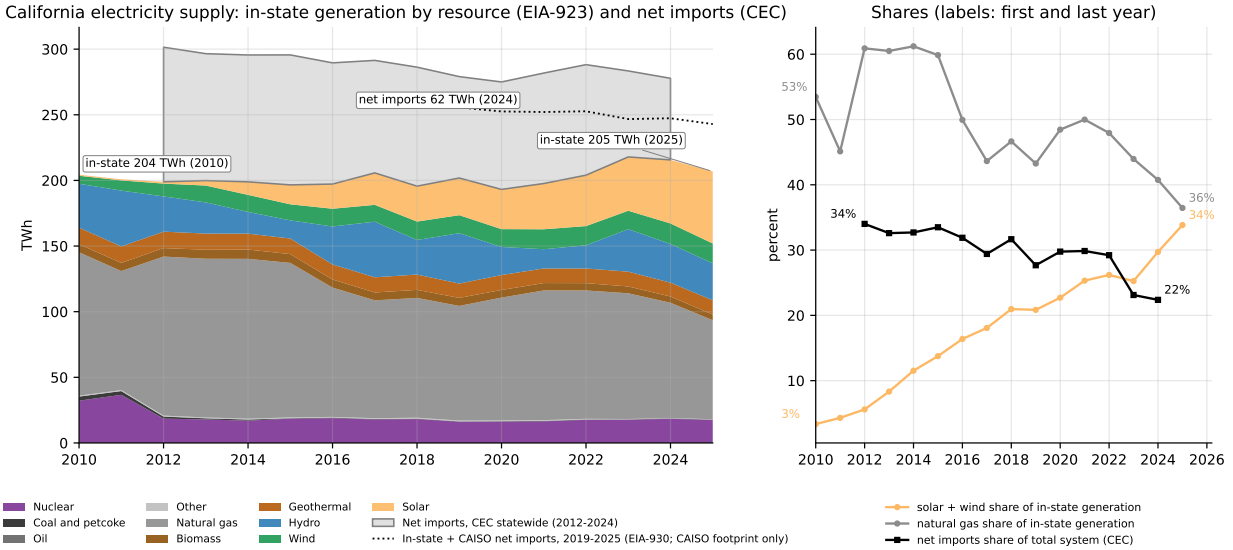


Figure 4.1: California electricity supply, 2010–2025: in-state generation by resource (EIA-923) stacked with CEC statewide net imports (2012–2024), the CAISO-only net imports of Chapter 2 as a proxy, and the shares of solar plus wind, natural gas and imports.

Table 4.1: In-state generation by resource and net imports, selected years (TWh). Batteries and pumped storage are net of charging. Source: EIA-923, CEC Energy Almanac, EIA-930.

Resource (TWh)	2010	2015	2020	2023	2024	2025
Natural gas	109.2	117.7	93.6	95.8	87.9	75.4
Nuclear	32.2	18.5	16.3	17.7	18.4	17.6
Hydro	33.4	13.8	21.4	32.4	29.5	28.1
Geothermal	12.6	11.9	11.4	11.3	10.5	10.9
Biomass	6.2	6.7	5.8	5.2	4.8	4.3
Wind	6.1	12.2	13.6	14.0	15.6	15.1
Solar	0.8	14.8	30.3	41.0	48.4	54.9
Coal and petcoke	3.1	0.3	0.3	0.2	0.2	0.2
Oil	0.1	0.1	0.0	0.0	0.0	0.0
Other	0.7	0.6	0.6	0.3	0.2	0.3
Batteries	0.0	-0.0	-0.0	-0.9	-1.4	-1.9
Pumped storage	-0.2	0.1	-0.0	-0.4	-0.0	0.2
Total in-state	204.1	196.7	193.1	216.6	214.2	205.1
Net imports, CEC statewide	–	99.0	81.9	65.5	62.2	–
Net imports, CAISO only (EIA-930)	–	–	59.4	28.8	31.7	36.1

In-state generation was 204 TWh in 2010 and 205 TWh in 2025, but its composition inverted (Figure 4.1, Table 4.1). Natural gas fell from 109 to 75 TWh, from 53 to 36 percent of in-state generation, and solar rose from under 1 to 55 TWh, so solar and wind together went from 3 to 34 percent. Nuclear halved with the closure of San Onofre, and hydro swung between 14 TWh in the 2015 drought and 42 TWh in 2017. Statewide net imports fell from 103 TWh in 2012, a third of the total system, to 62 TWh in 2024, 22 percent. The total system therefore shrank from 302 to 278 TWh between 2012 and 2024 while in-state generation grew, which is the same flat-demand picture that Chapter 2 found at the CAISO level, now with the import share falling.

4.2 Capacity, storage and curtailment

Operable nameplate capacity by technology comes from the EIA-860 annual generator files, classified the same way [U.S. Energy Information Administration, 2026a]; battery energy capacity comes from the EIA-860 energy storage file, which begins in 2016; curtailment is CAISO’s monthly wind and solar curtailment series [California Independent System Operator, 2026a].

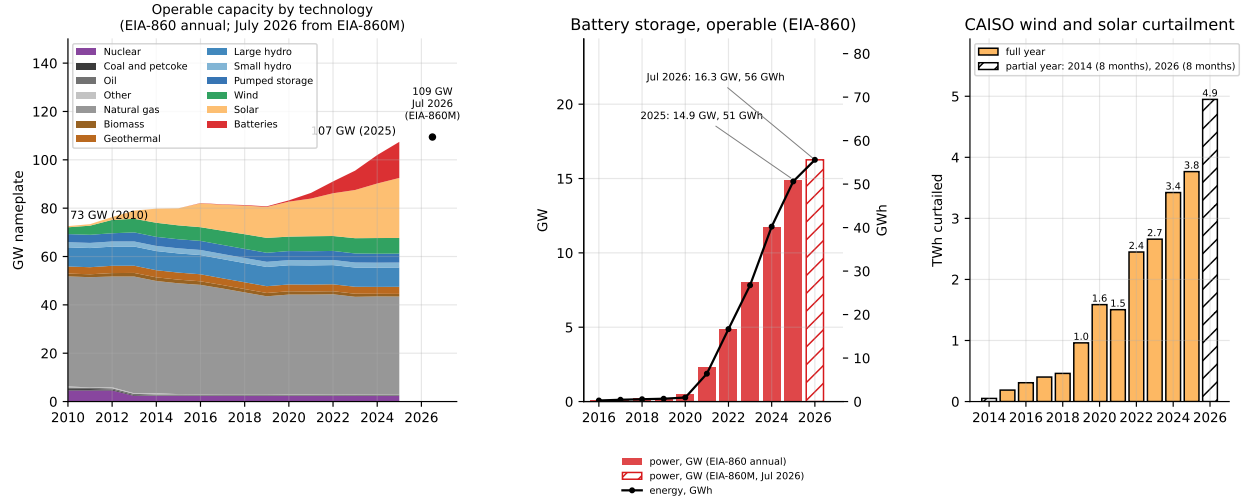


Figure 4.2: Operable capacity by technology in California (EIA-860, left), battery power and energy capacity (EIA-860 energy storage file, middle) and CAISO wind and solar curtailment by year (right; 2026 covers January to August).

Table 4.2: Operable nameplate capacity by technology, battery energy capacity and CAISO curtailment, selected years and the July 2026 inventory. Source: EIA-860, EIA-860M, CAISO.

Technology (GW nameplate)	2010	2015	2020	2023	2024	2025	Jul 2026
Natural gas	45.5	45.8	41.2	40.4	40.5	40.4	40.4
Nuclear	4.6	2.3	2.3	2.3	2.3	2.3	2.3
Large hydro	7.9	7.9	7.9	7.9	8.0	8.0	8.0
Small hydro	2.1	2.1	2.1	2.1	2.0	2.0	2.0
Pumped storage	3.4	3.8	3.7	3.7	3.7	3.7	3.7
Geothermal	2.9	2.9	2.8	2.9	2.9	2.9	2.9
Biomass	1.2	1.5	1.4	1.3	1.1	1.1	1.1
Wind	2.8	5.7	6.0	6.3	6.5	6.5	6.5
Solar	0.5	7.1	14.5	19.9	22.6	24.8	25.5
Batteries	–	0.0	0.5	8.0	11.8	14.9	16.3
Coal and petcoke	0.6	0.1	0.1	0.1	0.1	0.1	0.1
Oil	0.7	0.5	0.5	0.5	0.5	0.6	0.6
Other	0.4	0.1	0.1	0.1	0.1	0.1	0.1
Total	72.6	79.9	83.2	95.5	102.0	107.4	109.4
Battery energy capacity (GWh)	–	–	0.9	26.7	40.2	50.6	55.6
CAISO wind and solar curtailment (TWh)	–	0.19	1.59	2.66	3.42	3.77	4.95 (Jan–Aug)

The 2010–2025 columns are the EIA-860 annual files; the July 2026 column is the EIA-860M monthly inventory, which is preliminary.

Capacity rose from 73 GW in 2010 to 107 GW in 2025 (Figure 4.2, Table 4.2). Solar added

24 GW and batteries 15 GW, the latter almost entirely since 2020 and holding 51 GWh of energy across 281 units by 2025; gas capacity fell only from 45 to 40 GW and remains the largest block. The battery fleet is what moved the net peak later in the evening in Chapter 2. Curtailment of wind and solar rose from 0.2 TWh in 2015 to 3.8 TWh in 2025, and reached 4.9 TWh in the first eight months of 2026 alone, so the midday surplus that Chapter 2 measured in prices is now also visible as discarded energy of the order of the output of a gigawatt of solar.

The most recent data. The CEC has not yet published 2025 statewide imports and the EIA-860 annual file ends in 2025, so the latest observations come from three other files, each preliminary (Table 4.3). The EIA-923 monthly file through June 2026, which includes EIA’s state-level increment rows for the plants that report only annually, puts January to June in-state generation at 89.6 TWh against 97.4 TWh in the same months of 2025, with natural gas down 28 percent, hydro down 9 percent and solar up 6 percent; CAISO net imports over the same months rose from 15.7 to 20.6 TWh. The EIA-860M inventory of July 2026 puts operable capacity at 109 GW, with 16.3 GW and 56 GWh of batteries in 305 units, and CAISO curtailment through August 2026 had already exceeded the whole of 2025. The EIA-860M file for August 2026 had not been published when this chapter was built.

Table 4.3: January to June generation by resource, 2025 final against 2026 preliminary, and CAISO net imports over the same months. Source: EIA-923 annual 2025 and monthly June 2026 files, EIA-930.

January–June (TWh)	2025 (final)	2026 (preliminary)	Change (%)
Natural gas	30.6	22.1	-28
Nuclear	8.7	9.5	+9
Hydro	15.1	13.7	-9
Geothermal	5.4	5.2	-3
Biomass	2.1	2.1	+2
Wind	8.6	8.8	+2
Solar	27.5	29.1	+6
Coal and petcoke	0.1	0.0	-88
Oil	0.0	0.0	–
Other	0.1	0.1	+34
Batteries	-0.9	-1.1	+15
Pumped storage	-0.0	-0.2	–
Total in-state	97.4	89.6	-8
of which EIA state-level increment rows	0.2	24.8	–
CAISO net imports, EIA-930 (CAISO footprint)	15.7	20.6	+31

4.3 Retirement schedule

Planned retirements come from the EIA-860M operating list of July 2026, which carries a planned retirement date for each unit that has reported one [U.S. Energy Information Administration, 2026b], overlaid with the compliance schedule of the State Water Board’s once-through-cooling policy as amended in August 2023 [State Water Resources Control Board, 2023]. The two agree where both exist: Alamitos units 3 to 5 and Huntington Beach unit 2 report December 2026 and the policy requires compliance by December 31, 2026; Ormond Beach reports January 2027 against the same policy date; Scattergood units 1 and 2 report 2029 against December 31, 2029. Three cases carry no EIA date: LADWP’s Haynes units 1, 2 and 8 and Harbor unit 5, which the policy requires

to comply by December 31, 2029, and Diablo Canyon, whose compliance date SB 846 extended to October 31, 2030 and whose continued operation beyond that depends on federal licence renewal and further state action. The policy dates are used for those units in case C below, and Diablo Canyon is carried as a scenario.

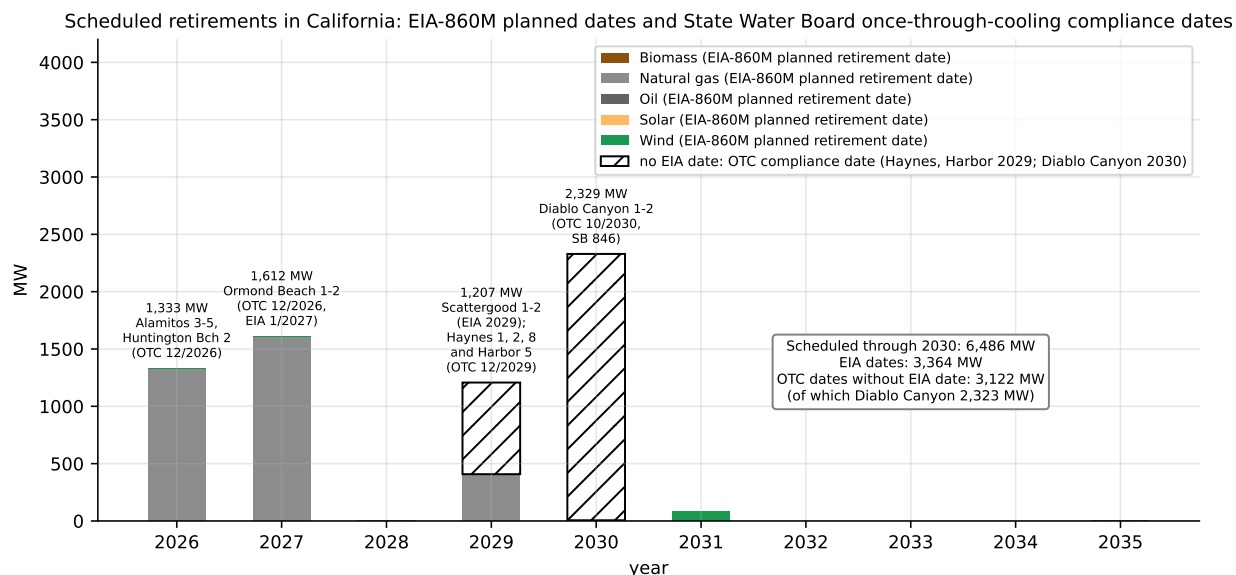


Figure 4.3: Scheduled retirements in California by year: EIA-860M planned retirement dates by technology, with the once-through-cooling compliance dates hatched for the units that report no EIA date.

Table 4.4: Once-through-cooling units and other scheduled retirements through 2030. Source: EIA-860M July 2026, State Water Board OTC Policy Table 1.

Plant	Units	Technology	MW	In service	OTC compliance date	EIA-860M planned retirement
AES Alamitos LLC	3, 4, 5	Natural Gas Steam Turbine	1,115	1961–1964	2026-12-31	2026
AES Huntington Beach LLC	2	Natural Gas Steam Turbine	218	1958–1958	2026-12-31	2026
Ormond Beach	1, 2	Natural Gas Steam Turbine	1,612	1971–1973	2026-12-31	2027
Haynes	1, 2, 8	Natural Gas Steam Turbine	724	1962–2005	2029-12-31	none
Harbor	5	Natural Gas Fired Combined Cycle	75	1995–1995	2029-12-31	none
Scattergood	1, 2	Natural Gas Steam Turbine	326	1958–1959	2029-12-31	2029
Diablo Canyon	1, 2	Nuclear	2,323	1985–1986	2030-10-31	none
Other units with an EIA date through 2030	5 units	mixed	92	–	–	2028–2030

Scheduled retirements through 2030 total 6.5 GW: 3.4 GW with EIA dates, almost all of it the coastal gas steam units of 2026, 2027 and 2029, plus 0.8 GW of LADWP units at their policy dates and the 2.3 GW of Diablo Canyon. Beyond 2030 the EIA record holds almost nothing, which reflects how retirement dates are reported rather than the age of the fleet.

4.4 A supply-side realization model

The CEC discounts data center requests by a confidence level for each tier. The supply side deserves the same treatment, and the EIA-860M monthly files make it possible to estimate one from California's own record. For every California unit in the Planned sheet of the January file of each year from 2016 to 2022, the outcome by the December 2025 file is recorded: operating (or operated and since retired), cancelled (the file's cumulative Canceled or Postponed sheet), still planned, or dropped from the survey without a cancellation record, which is treated as cancelled. Units that appear in more than one January list contribute one observation per vintage, and standard errors are clustered by unit. The sample is 925 unit-vintages of 513 distinct units and 64 GW of planned capacity.

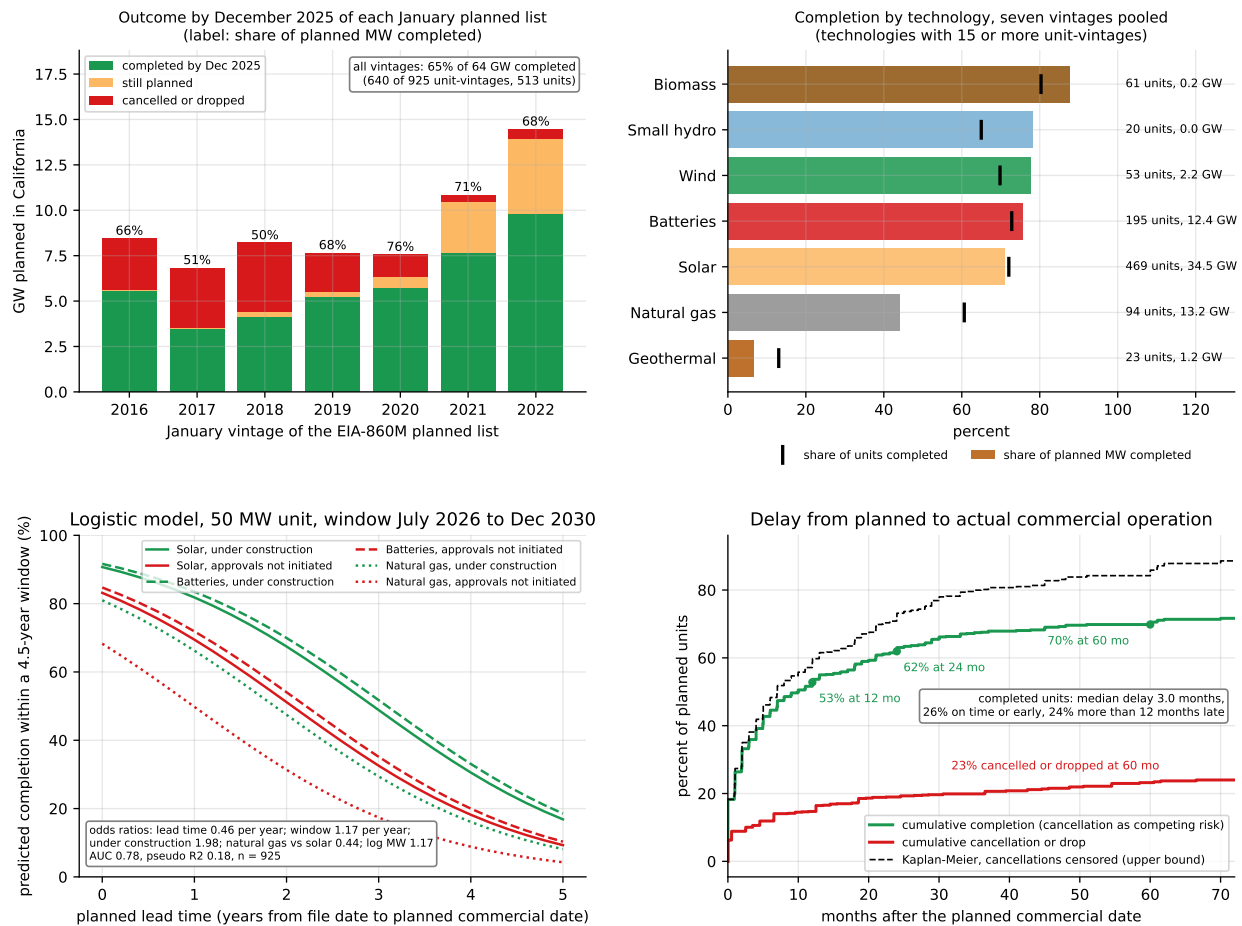


Figure 4.4: The realization model. Top left: outcome by December 2025 of each January planned list, capacity-weighted. Top right: completion by technology, seven vintages pooled. Bottom left: predicted completion within a 4.5-year window from the logistic model for a 50 MW unit. Bottom right: time from the planned to the actual commercial date, as Kaplan-Meier with cancellations censored and as cumulative incidence with cancellation as a competing risk.

Table 4.5: Outcome by December 2025 of the California planned lists of January 2016 to 2022, by vintage and by construction status at the vintage.

Vintage or status at the vintage	Unit-vintages	Planned MW	Completed (% units)	Completed (% MW)	Cancelled MW	Still planned MW	Median delay (months)
January 2016	145	8,454	77	66	2,842	20	2
January 2017	110	6,837	66	51	3,292	70	1
January 2018	110	8,209	61	50	3,801	270	2
January 2019	115	7,660	61	68	2,145	270	3
January 2020	127	7,569	72	76	1,245	556	6
January 2021	150	10,860	74	71	372	2,828	4
January 2022	168	14,466	69	68	519	4,146	4
Status: under construction	329	20,536	85	77	4,784	5	2
Status: approvals not initiated	202	16,685	55	54	2,312	5,407	15
Status: approvals pending	172	14,869	63	59	5,214	927	7
Status: approved, not started	150	10,582	53	66	1,798	1,817	5
Status: construction complete	72	1,383	81	92	108	3	0

Table 4.6: Outcome by technology, seven vintages pooled (technologies with at least ten unit-vintages).

Technology	Unit-vintages	Planned MW	Completed (% units)	Completed (% MW)	Cancelled MW	Still planned MW	Median delay (months)
Solar	469	34,524	72	71	5,130	4,878	3
Natural gas	94	13,164	61	44	7,379	0	1
Batteries	195	12,434	73	76	343	2,678	4
Wind	53	2,204	70	78	195	298	5
Geothermal	23	1,158	13	6	1,083	0	7
Biomass	61	164	80	88	15	5	2
Small hydro	20	45	65	78	10	0	28

Across the seven vintages, 69 percent of units and 65 percent of planned megawatts were operating by December 2025 (Table 4.5); by vintage the capacity-weighted completion rate ranges from 50 percent for January 2018 to 76 percent for January 2020, with the 2021 and 2022 lists still carrying 3 to 4 GW of unresolved projects. Technology matters less than one might expect among the resources that dominate the list: solar completed 71 percent of its planned megawatts, batteries 76 and wind 78 (Table 4.6). Natural gas completed 44 percent and geothermal 6 percent, the latter on 23 unit-vintages. Construction status at the vintage is the strongest single signal: units already under construction completed 77 percent of their megawatts, units whose approvals had not been initiated 54 percent.

Table 4.7: Logistic model of completion by December 2025 (unit-vintage observations; standard errors clustered by unit; reference categories solar and approvals not initiated).

Term	Coefficient	Cluster SE	Odds ratio	<i>p</i>
Intercept (solar, approvals not initiated)	0.290	0.554	1.34	0.601
Technology: Batteries	0.116	0.348	1.12	0.739
Technology: Biomass	-0.172	0.654	0.84	0.793
Technology: Geothermal	-2.180	1.122	0.11	0.052
Technology: Natural gas	-0.830	0.466	0.44	0.075
Technology: Other	-2.489	1.169	0.08	0.033
Technology: Small hydro	-0.034	0.961	0.97	0.972
Technology: Wind	-0.094	0.899	0.91	0.917
Status: approvals pending	0.087	0.416	1.09	0.834
Status: approved, not started	-0.273	0.344	0.76	0.429
Status: construction complete	0.257	0.491	1.29	0.601
Status: under construction	0.682	0.355	1.98	0.055
log nameplate MW	0.154	0.092	1.17	0.094
Planned lead time (years)	-0.775	0.189	0.46	<0.001
Observation window (years)	0.158	0.056	1.17	0.005
Observations	925 unit-vintages (513 units)			
Pseudo R^2 / in-sample AUC	0.178 / 0.778			

Table 4.8: Three specifications of the completion model: the main model, the model without construction status, and the literal specification with planned year and vintage fixed effects.

Specification	<i>n</i>	Pseudo R^2	AUC	OR: lead time (A, B) or planned year (C), per year	OR: under construction	OR: gas vs solar	OR: log MW
A. lead time + window + status (main)	925	0.178	0.778	0.46	1.98	0.44	1.17
B. lead time + window, no status	925	0.166	0.774	0.40	–	0.56	1.18
C. planned year + vintage fixed effects + status	925	0.162	0.764	0.55	2.60	0.45	1.11

Table 4.9: Delay from the planned to the actual commercial date, seven vintages pooled.

Statistic	Value
Completed unit-vintages	640
Median delay, planned to actual commercial date	3.0 months
Mean delay	7.4 months
On time or early	26%
More than 12 months late	24%
90th percentile delay	24 months
Cumulative completion by 12 months after the planned date (cancellation as competing risk)	53%
Cumulative completion by 24 months after the planned date (cancellation as competing risk)	62%
Cumulative completion by 36 months after the planned date (cancellation as competing risk)	68%
Cumulative completion by 60 months after the planned date (cancellation as competing risk)	70%
Cumulative cancellation or drop by 60 months	23%

The logistic model (Table 4.7) puts numbers on these patterns while holding the others fixed. Each additional year of planned lead time cuts the odds of completion within the window by more than half (odds ratio 0.46, $p < 0.001$); each additional year of observation raises them by 17

percent ($p = 0.005$), which is the mechanical effect of giving late projects time to finish; being under construction at the vintage doubles the odds ($p = 0.055$); gas and geothermal have lower odds than solar at conventional or near-conventional significance, and size adds a little (odds ratio 1.17 per log-unit of megawatts, $p = 0.09$). The in-sample AUC is 0.78. Lead time and observation window are a reparametrisation of the planned year and the vintage; the literal specification with planned year and vintage fixed effects (Table 4.8) gives the same ranking of technologies and construction status. The delay curves (Table 4.9) describe the projects that do finish and the ones that do not: among completed units the median delay from the planned to the actual commercial date is 3 months, a quarter are on time or early and a quarter are more than a year late. Treating cancellation as a competing risk, 53 percent of planned units are operating within 12 months of their planned date, 62 percent within 24 and 70 percent within 60, while 23 percent have been cancelled or dropped by then; the Kaplan-Meier curve with cancellations censored, an upper bound, reaches about 88 percent at six years.

4.5 Probability-weighted supply for 2030

The July 2026 EIA-860M planned list for California holds 229 units and 20.1 GW with planned commercial dates through 2030: 9.9 GW of batteries, 7.9 GW of solar, 1.6 GW in two pumped-storage projects whose approvals have not been initiated, 0.3 GW of wind and 0.2 GW of gas engines. Each unit is weighted by the model’s completion probability for a window ending December 2030, given its technology, size, lead time and construction status. The weighted list is 13.2 GW, two thirds of the nominal one: 71 percent of the battery and solar megawatts survive, 78 percent of wind, 31 percent of gas and 13 percent of the pumped storage. By status, 88 percent of the megawatts already under construction survive against 36 percent of those whose approvals have not been initiated.

Table 4.10: ELCC values and derates used for peak contribution, with realised 2023–2025 capacity factors used for energy. Source: CPUC/E3-Astrape 2023 MTR ELCC study, EIA-923 and EIA-860; assumptions as stated.

Technology	ELCC or derate	Capacity factor 2023–2025	Basis
Natural gas	0.950	0.244	assumption: firm resource net of forced-outage and ambient derate
Nuclear	0.950	0.879	assumption: firm resource net of forced-outage and ambient derate
Large hydro	0.550	0.342	assumption: summer-peak availability of hydro (CAISO NQC uses historical exceedance)
Small hydro	0.450	0.342	assumption: summer-peak availability of hydro (CAISO NQC uses historical exceedance)
Geothermal	0.950	0.430	assumption: firm resource net of forced-outage and ambient derate
Biomass	0.900	0.470	assumption: firm resource net of forced-outage and ambient derate
Wind	0.147	0.265	CPUC/E3-Astrape 2023 MTR ELCC study, Table 1, Tranche 6 (2028): 4-hour battery, 8-hour PSH, utility solar, in-state wind
Solar	0.088	0.244	CPUC/E3-Astrape 2023 MTR ELCC study, Table 1, Tranche 6 (2028): 4-hour battery, 8-hour PSH, utility solar, in-state wind
Batteries	0.765	-0.014	CPUC/E3-Astrape 2023 MTR ELCC study, Table 1, Tranche 6 (2028): 4-hour battery, 8-hour PSH, utility solar, in-state wind
Pumped storage	0.887	-0.002	CPUC/E3-Astrape 2023 MTR ELCC study, Table 1, Tranche 6 (2028): 4-hour battery, 8-hour PSH, utility solar, in-state wind
Coal and petcoke	0.950	0.419	assumption: firm resource net of forced-outage and ambient derate
Oil	0.900	0.009	assumption: firm resource net of forced-outage and ambient derate
Other	0.800	0.298	assumption: firm resource net of forced-outage and ambient derate

Energy is capacity times the realised 2023 to 2025 capacity factor of each class (batteries and pumped storage contribute no net energy, and for gas the fleet-average factor overstates the energy of the peaking units that retire). Peak contribution uses effective load carrying capability: the CPUC’s 2028 tranche values from the E3 and Astrape mid-term reliability study for solar (8.8 percent), in-state wind (14.7), four-hour batteries (76.5) and pumped storage (88.7) [Energy and Environmental Economics and Astrapé Consulting, 2023, California Public Utilities Commission, Energy Division, 2023], and stated derates for firm resources and hydro (Table 4.10). Three cases are reported (Figure 4.5, Table 4.11): everything on the list builds; the model-weighted list; and the model-weighted list minus the retirements of Section 4.3, with Diablo Canyon at its October 2030 policy date and a sensitivity in which it continues.

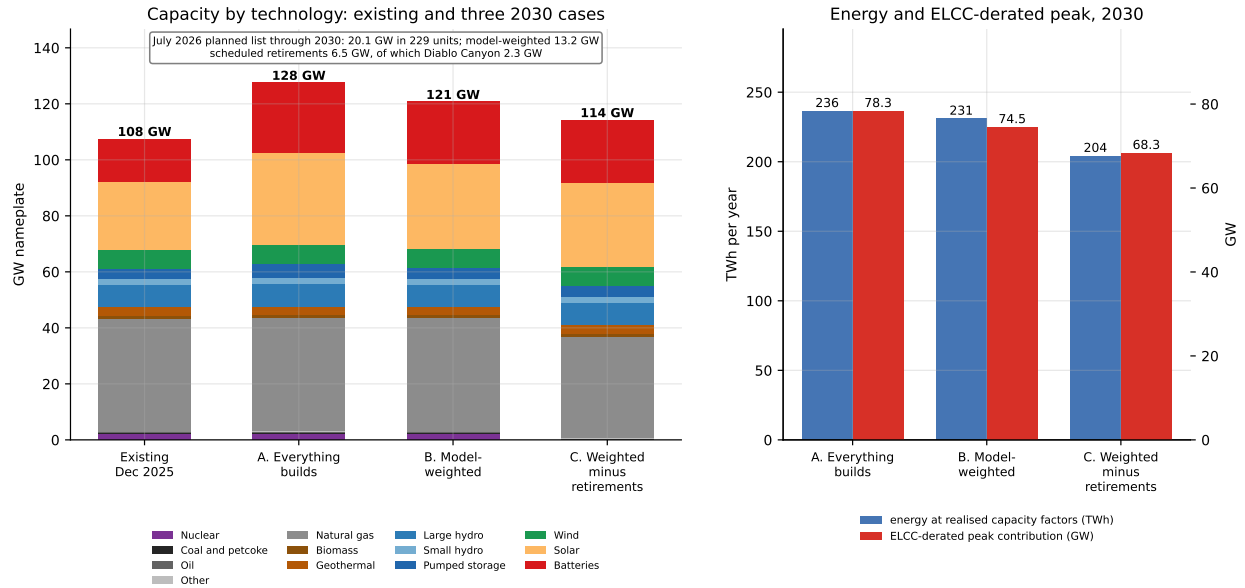


Figure 4.5: Capacity by technology for the existing fleet and the three 2030 cases (left), and energy at realised capacity factors and ELCC-derated peak contribution by case (right).

Table 4.11: Planned supply for 2030 in three cases: capacity by technology (MW), energy at realised capacity factors and ELCC-derated peak contribution.

Technology	Existing Dec 2025	A. Everything builds	B. Model-weighted	C. Weighted minus retirements
Natural gas	40,414	40,585	40,467	36,308
Nuclear	2,323	2,323	2,323	0
Large hydro	7,979	7,979	7,979	7,979
Small hydro	2,041	2,050	2,047	2,047
Pumped storage	3,704	5,304	3,917	3,917
Geothermal	2,868	2,868	2,868	2,868
Biomass	1,107	1,136	1,127	1,124
Wind	6,530	6,874	6,799	6,799
Solar	24,625	32,556	30,274	30,274
Batteries	15,216	25,137	22,225	22,225
Coal and petcoke	62	62	62	62
Oil	540	540	540	538
Other	103	153	104	104
Total capacity (MW)	107,512	127,566	120,732	114,246
Energy at realised capacity factors (TWh)	–	236	231	204
ELCC-derated peak contribution (MW)	–	78,297	74,467	68,306
Case C with Diablo Canyon continuing: capacity / energy / peak	–	–	–	116,569 MW / 222 TWh / 70,512 MW

If everything builds, California’s nameplate capacity reaches 128 GW in 2030, 236 TWh a year at realised capacity factors and 78 GW of ELCC-derated peak contribution. The model-weighted case gives 121 GW, 231 TWh and 74 GW. Subtracting 6.5 GW of retirements leaves 114 GW, 204 TWh and 68 GW, or 117 GW, 222 TWh and 71 GW if Diablo Canyon continues. Net of retirements, the supply side therefore adds about 7 to 9 GW of nameplate and 0 to 3 GW of ELCC-derated peak contribution by 2030 relative to the December 2025 fleet, and its energy falls if Diablo Canyon closes, because almost everything being added is solar and batteries whose peak value is a fraction of nameplate. Against that stand the 4 GW of managed net peak growth in the adopted forecast and the 20.7 GW of data center requests located in California from Chapter 3. Setting the two probability-weighted sides against each other is the subject of the next chapter.

Chapter 5

Putting It Together

Chapter 6

Discussion

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